

Where the Seconds Go: Supplement

The prediction scorecard

Every prediction committed before a GPU job started, split into its separable clauses and scored under the rule of Appendix B of the paper. Jobs 073–098: generated by `scripts/scorecard.py` from `prereg/scorecard_clauses.json`. Job 099, the host panel: scored by machine by `scripts/panel_099.py` (`prereg/scorecard_099.json`); outcome notes in `prereg/panel_outcome_099.md`.

Table 1: Prediction scorecard, jobs 073–098: every committed prediction split into its separable clauses and scored under one rule. *Held*: the point estimate is on the predicted side and the 95% paired-bootstrap interval excludes the threshold (for a band, lies inside it). *Held (point)*: the point estimate is on the predicted side but the interval includes the threshold, or no interval exists. *Untested*: the clause could not be evaluated. *Void*: the job was voided (084, a throttled card; its predictions are scored on 084b). Types: sign (a direction), threshold (a numeric bar), band (an interval), equality (a deterministic outcome). Measured values are ratios of mean speeds with 95% intervals unless the clause says otherwise; `prereg/scorecard_clauses.json` names the source of every number. Totals: 565 clauses, 218 held, 193 held (point), 142 failed, 8 untested, 4 void.

Job	Clause	Type	Measured	Status
Jobs 073–075: 20 clauses, 7 held, 3 held (point), 9 failed, 1 untested, 0 void				
073 P1a	ours+FETCH $\geq$ 5% ahead of FreeToken, 11%	threshold	1.109 [1.098, 1.120]	held
073 P1b	ours+FETCH $\geq$ 5% ahead of FreeToken, 25%	threshold	1.084 [1.076, 1.092]	held
073 P1c	ours+FETCH within $\pm 5\%$ of FreeToken, 40%	band	0.941 [0.933, 0.951]	failed
073 P2	ours+FETCH $\geq 1.8\times$ llama.cpp, every budget	threshold	1.832 [1.808, 1.856]; 2.358 [2.322, 2.396]; 2.647 [2.605, 2.686]	held
073 P3a	same-problem warm-up reads 5-15% faster, the two caches	band	0.991 [0.962, 1.028]; 0.997 [0.986, 1.008]	failed
073 P3b	same-problem warm-up within $\pm 2\%$, llama.cpp	band	0.962 [0.934, 0.979]	failed
073 P4	$\geq 80\%$ of greedy outputs identical to llama.cpp's, every system	equality	ours (6 configurations): 0.000-0.033; FreeToken (6 configurations): 0.000; llama.cpp -ncmoe 32 / 22: 0.700 / 0.400	failed
074 P1	ec-bench launch time ≤ 0.25 ms, C51	threshold	0.201	held (point)
074 P2a	ours v2 ≥ 127 tok/s, C51	threshold	95.1	failed
074 P2b	ours v2 ahead of FreeToken offload, C51	sign	0.831 [0.817, 0.845]	failed
074 P3a	ours v2 ≥ 59 tok/s, C14	threshold	41.5	failed
074 P3b	ours v2 ≥ 91 tok/s, C32	threshold	65.3	failed
074 P4	half-life 64 within $\pm 2\%$ of half-life 16, C51	band	1.012 [0.997, 1.027]	held (point)
074b P1	GPU sampling cuts between-step host time by ≥ 1 ms	threshold	1.6	held (point)
074b P2a	ours v2 $\geq 10\%$ faster with GPU sampling, C51	threshold	1.117 [1.109, 1.125]	held
074b P2b	stock llama.cpp $\geq 3\%$ faster with GPU sampling	threshold	–	<i>untested</i>
074b P3	ours v2+FETCH with GPU sampling $\geq$ FreeToken offload, C51	sign	1.074 [1.063, 1.085]	held
075 P1	ours ahead of FreeToken at 40% (RTX PRO 6000)	sign	1.122 [1.107, 1.137]	held
075 P2	ours within -3 to +10% of FreeToken at 60%	band	1.118 [1.109, 1.127]	failed
075 P3	ours at 60% $\geq 55\%$ of the all-in-VRAM speed	threshold	0.726 [0.717, 0.735]	held
Jobs 076–081: 23 clauses, 19 held, 4 held (point), 0 failed, 0 untested, 0 void				
076 P1	ours ahead of FreeToken at 11/25/40%	sign	1.205 [1.190, 1.220]; 1.217 [1.201, 1.234]; 1.118 [1.108, 1.129]	held
076 P2	ours $\geq 1.8\times$ llama.cpp, every budget	threshold	1.924 [1.894, 1.956]; 2.683 [2.631, 2.735]; 3.191 [3.130, 3.250]	held
076 P3	the law within 8% on own-text runs (four configurations)	band	1.009 [0.962, 1.066]; 1.006 [0.947, 1.086]; 1.052 [0.960, 1.164]; 0.960 [0.909, 1.024]	held (point)
077 P1a	both entrants run end to end on gpt-oss-120b	equality	30 of 30 problems at every budget for both	held

continued

Job	Clause	Type	Measured	Status
077 P1b	both entrants faster than llama.cpp at 25/40%	sign	pipeshard 25 / 40%: 1.348 [1.336, 1.360]; 1.417 [1.399, 1.435]; leloch 25 / 40%: 1.446 [1.431, 1.461]; 1.280 [1.269, 1.291]	held
077 P2	ours ahead of both entrants, every budget	sign	1.421 [1.403, 1.439]; 1.896 [1.871, 1.921]; 2.343 [2.293, 2.391]	held
078 P1	ours ahead of FreeToken at 12.5/25% (Qwen3)	sign	1.071 [1.063, 1.080]; 1.067 [1.043, 1.094]	held
078 P2	ours ahead of KTransformers, every budget	sign	2.168 [2.140, 2.197]; 3.059 [2.975, 3.162]; 4.420 [4.310, 4.550]	held
078 P3	ours $\geq 1.5\times$ llama.cpp, every budget	threshold	2.122 [2.092, 2.151]; 3.051 [2.967, 3.152]; 4.470 [4.350, 4.609]	held
079 P1a	host-side time per step < 1 ms	threshold	0.266	held (point)
079 P1b	host-side time < 10% of the token time	threshold	0.027	held (point)
079 P2	$\geq 1.5\times$ FreeToken’s kernels per token	threshold	1.606	held (point)
079 P3	ours within $\pm 5\%$ of FreeToken at 43.75%	band	0.991 [0.975, 1.009]	held
080 P1	law table $\geq 3\%$ over the current table, 43.75%	threshold	1.048 [1.041, 1.055]	held
080 P2	law table within 2% of the sweep’s best, 43.75%	band	0.991 [0.986, 0.996]	held
080 P3	best table leads FreeToken, 43.75%	sign	1.049 [1.036, 1.064]	held
080 P4	law table beats the current table, 25%	sign	1.067 [1.062, 1.071]	held
081 P1	law table beats current, gpt-oss, every budget	sign	1.079 [1.072, 1.085]; 1.054 [1.041, 1.066]; 1.033 [1.021, 1.044]	held
081 P2	ours leads FreeToken, gpt-oss 11/25/40%	sign	1.294 [1.278, 1.312]; 1.275 [1.254, 1.295]; 1.154 [1.134, 1.172]	held
081 P3a	law table beats current, Qwen3 12.5%	sign	1.080 [1.073, 1.084]	held
081 P3b	ours leads FreeToken, Qwen3 12.5%	sign	1.032 [1.022, 1.042]	held
081 P4a	ours $\geq 1.8\times$ llama.cpp, gpt-oss, every budget	threshold	2.001 [1.964, 2.041]; 2.731 [2.668, 2.795]; 3.196 [3.133, 3.259]	held
081 P4b	ours $\geq 2\times$ llama.cpp, Qwen3 12.5%	threshold	2.051 [2.026, 2.077]	held
Jobs 082–087: 43 clauses, 24 held, 4 held (point), 10 failed, 1 untested, 4 void				
082 P1	steady-state lead over FreeToken, gpt-oss, held-out prompts	sign	–	untested
082 P2	FreeToken gains more after 256 tokens than ours, every budget	sign	FreeToken: 1.017 / 1.065 / 1.071; ours: 1.038 / 1.094 / 1.117	failed
082 P3	FreeToken ahead in steady state, Qwen3 43.75%	sign	1.054 [1.025, 1.086]	failed
082 P4a	every ablation step $\geq -1\%$	threshold	0.723 [0.712, 0.733]; 0.744 [0.740, 0.748]	failed
082 P4b	ours $\geq 2\times$ stock at 25%, both models	threshold	2.658 [2.566, 2.749]; 2.800 [2.743, 2.858]	held
082 P5a	parity: $\geq 98\%$ top-1 agreement, both models	threshold	0.986 / 0.993	held
082 P5b	parity: $ \Delta\text{NLL} \leq 1\%$, both models	threshold	-0.0022 / 0.0031	held
082 P6	llama.cpp Qwen3 25/43.75% within 5% of the law	band	0.966 [0.962, 0.969]; 0.962 [0.958, 0.965]	held
083 P1	still reasoning at 2,048 tokens on ≥ 8 of 10, both systems	threshold	ours: 9 / 9 / 10; FreeToken: 10 / 9 / 10	held
083 P2	ours within 15% of its Table 1 rate	band	1.046 [1.019, 1.072]; 1.037 [1.013, 1.060]; 1.033 [1.003, 1.061]	held
083 P3a	ours leads FreeToken, all tokens, 11/25/40%	sign	1.358 [1.338, 1.378]; 1.322 [1.303, 1.342]; 1.191 [1.172, 1.212]	held
083 P3b	ours leads FreeToken, tokens 257-2,048	sign	1.366 [1.343, 1.389]; 1.328 [1.310, 1.350]; 1.194 [1.175, 1.215]	held
083 P4	lead $\geq 15\%$ at 11 and 25%	threshold	1.358 [1.338, 1.378]; 1.322 [1.303, 1.342]	held
084 P1	Qwen3 all in VRAM at 120-190 tok/s	band	–	void
084 P2	gpt-oss all in VRAM within 5% of 261.4 tok/s	band	–	void
084 P3	ours with 128 slots within 5% of all in VRAM, both	band	–	void
084 P4	sim. hit rate within 5 points; ours at 20-50% of the limit	band	–	void
084b P1	Qwen3 all in VRAM at 120-190 tok/s	band	177.8 [177.1, 178.4]	held
084b P2	gpt-oss all in VRAM within 5% of 261.4 tok/s	band	0.977 [0.971, 0.982]	held
084b P3	ours with 128 slots within 5% of all in VRAM, both	band	0.997 [0.991, 1.004]; 1.047 [1.043, 1.052]	held (point)
084b P4a	sim. hit rate within 5 points of the engine’s, every budget	band	-0.014 / -0.008 / -0.0003 / -0.053 / -0.026 / -0.006	failed
084b P4b	ours at 20-50% of the speed limit, every budget	band	0.413 / 0.256 / 0.294 / 0.433 / 0.302 / 0.352	held (point)
084c	<i>No predictions of its own: it supplies the gpt-oss routing trace that 084b-P4a and 084b-P4b are scored on.</i>	–	–	–
085 P1	law’s tables fetch less than on the headline machine	equality	gpt-oss: 0,0,0,1,1 (two entries lower); Qwen3: 0,0,0,1,1,2,2,3,3 (six entries lower)	held

continued

Job	Clause	Type	Measured	Status
085 P2a	law beats fixed, gpt-oss, every budget	sign	1.340 [1.327, 1.355]; 1.274 [1.256, 1.292]; 1.211 [1.187, 1.235]	held
085 P2b	law's gain larger than headline at ≥ 2 of 3 (pro rata of 4 of 6)	threshold	3 of 3 (every interval excludes the headline value); pooled with 085b: 6 of 6	held
085 P3	ours leads FreeToken, gpt-oss, every budget	sign	2.030 [1.993, 2.068]; 1.699 [1.662, 1.737]; 1.428 [1.391, 1.463]	held
085 P4	lead larger than headline at ≥ 2 of 3 (pro rata of 4 of 6)	threshold	3 of 3 (every interval excludes the headline value); pooled with 085b (1 of 3): 4 of 6	held
085 P5	llama.cpp at 25% within 5% of the law, gpt-oss	band	0.818 [0.817, 0.819]	failed
085b P1	law's table fetches less than on the headline machine	equality	0,0,1,1,2,2,3,3,4 (three entries lower)	held
085b P2a	law beats fixed, Qwen3, every budget	sign	1.184 [1.177, 1.190]; 1.149 [1.142, 1.157]; 1.094 [1.086, 1.101]	held
085b P2b	law's gain larger than headline at ≥ 2 of 3	threshold	3 of 3 (every interval excludes the headline value)	held
085b P3	ours leads FreeToken, Qwen3, every budget	sign	1.067 [1.058, 1.076]; 1.076 [1.063, 1.089]; 1.038 [1.017, 1.060]	held
085b P4	lead larger than headline at ≥ 2 of 3	threshold	1 of 3: 12.5% 1.067 [1.058, 1.076] > 1.032; 25% 1.076 [1.063, 1.089] < 1.153; 43.75% 1.038 [1.017, 1.060] below 1.049 with the interval covering it	failed
085b P5	llama.cpp at 25% within 5% of the law, Qwen3	band	0.974 [0.972, 0.976]	held
086 P1	FreeToken as shipped within 15% of its paper's rate	band	93.8 [91.3, 96.5]	held (point)
086 P2	ours leads FreeToken at 12.5/25/37.5%	sign	1.030 [1.023, 1.038]; 1.010 [0.998, 1.021]; 1.003 [0.994, 1.013]	failed
086 P3	ours $\geq 2\times$ llama.cpp, every budget	threshold	1.976 [1.950, 2.005]; 2.311 [2.264, 2.361]; 2.481 [2.441, 2.521]	failed
086 P4	law beats fixed, every budget	sign	1.071 [1.068, 1.074]; 1.060 [1.057, 1.063]; 1.057 [1.052, 1.061]	held
087 P1	profiled static cache $\geq 1.3\times$ stock, both models	threshold	1.307 [1.236, 1.379]; 1.889 [1.803, 1.969]	held (point)
087 P2	replacement adds less than placement, both models	sign	0.268 [0.122, 0.434]; -0.765 [-0.898, -0.619]	failed
087 P3	decayed frequency beats the profiled static cache, both	sign	1.575 [1.497, 1.674]; 1.124 [1.070, 1.187]	held
087 P4a	every step after the static cache $\geq -1\%$	threshold	0.877 [0.835, 0.924]	failed
087 P4b	ours $\geq 2\times$ stock, both models	threshold	2.680 [2.595, 2.763]; 2.815 [2.754, 2.877]	held
Jobs 088–098: 479 clauses, 168 held, 182 held (point), 123 failed, 6 untested, 0 void				
088 P1	FreeToken hybrid on 8 threads $\geq 1.3\times$ job 085's hybrid, every budget	threshold	3.12 / 2.04 / 1.4	held (point)
088 P2	FreeToken's best improves on job 085's best at ≥ 2 of 3 budgets	threshold	1.52 / 1.29 / 1.08	held (point)
088 P3a	ours leads FreeToken's best, 11% (CI > 1)	sign	1.582 [1.564, 1.603]	held
088 P3b	ours leads FreeToken's best, 25% (CI > 1)	sign	1.519 [1.472, 1.559]	held
088 P3c	ours leads FreeToken's best, 40% (CI > 1)	sign	1.506 [1.476, 1.537]	held
088 P3d	ours / FreeToken between $1.10\times$ and $1.50\times$, 11%	band	1.582 [1.564, 1.603]	failed
088 P3e	ours / FreeToken between $1.10\times$ and $1.50\times$, 25%	band	1.519 [1.472, 1.559]	failed
088 P3f	ours / FreeToken between $1.10\times$ and $1.50\times$, 40%	band	1.506 [1.476, 1.537]	failed
088 P4a	llama.cpp -t 8 within 10% of the law	band	-9.4	held (point)
088 P4b	llama.cpp -t 8 faster than -t 24	sign	1.219 [1.213, 1.225]	held
088 P5	the law's table is 0,0,0,1,1 or fetches less	equality	0,0,0,1,1	held
089 P1a	the card reports a 14,001 MHz memory clock	equality	14001 MHz	held
089 P1b	device read $\geq 1,500$ GB/s (not throttled)	threshold	1694	held
089 P2a	ours / FreeToken within ± 0.06 of Table 1, gpt-oss 11%	band	-0.088 [-0.099, -0.075]	failed
089 P2b	ours / FreeToken within ± 0.06 of Table 1, gpt-oss 25%	band	-0.078 [-0.112, -0.051]	failed
089 P2c	ours / FreeToken within ± 0.06 of Table 1, gpt-oss 40%	band	-0.061 [-0.077, -0.043]	failed
089 P2d	ours / FreeToken within ± 0.06 of Table 1, Qwen3 12.5%	band	-0.006 [-0.014, 0.004]	held
089 P2e	ours / FreeToken within ± 0.06 of Table 1, Qwen3 25%	band	-0.105 [-0.116, -0.094]	failed
089 P2f	ours / FreeToken within ± 0.06 of Table 1, Qwen3 43.75%	band	-0.075 [-0.087, -0.062]	failed
089 P3a	ours' speed within $\pm 12\%$ of host B's, gpt-oss 11%	band	-17.1	failed
089 P3b	ours' speed within $\pm 12\%$ of host B's, gpt-oss 25%	band	-13.8	failed
089 P3c	ours' speed 5-20% below host B's, gpt-oss 40%	band	-12.4	held (point)

continued

Job	Clause	Type	Measured	Status
089 P3d	ours' speed within $\pm 12\%$ of host B's, Qwen3 12.5%	band	-16.0	failed
089 P3e	ours' speed within $\pm 12\%$ of host B's, Qwen3 25%	band	-14.6	failed
089 P3f	ours' speed 5-20% below host B's, Qwen3 43.75%	band	-11.5	held (point)
089 P4	ours-first and FreeToken-first orders agree within 3%, every cell	band	-1.0 / 0.2 / 0.7 / 0.3 / -0.2 / 0.1	held (point)
089 P5a	LRU 10-30% slower than decayed frequency, gpt-oss 11%	band	18.5 [17.7, 19.3]	held
089 P5b	LRU 10-30% slower than decayed frequency, gpt-oss 25%	band	12.8 [11.5, 14.1]	held
089 P5c	LRU 10-30% slower than decayed frequency, gpt-oss 40%	band	7.8 [6.5, 9.1]	failed
089 P5d	LRU 10-30% slower than decayed frequency, Qwen3 12.5%	band	20.9 [20.6, 21.3]	held
089 P5e	LRU 10-30% slower than decayed frequency, Qwen3 25%	band	15.4 [14.9, 15.9]	held
089 P5f	LRU 10-30% slower than decayed frequency, Qwen3 43.75%	band	6.4 [5.9, 6.9]	failed
089 P5g	LRU trails FreeToken at Qwen3 12.5%	sign	0.814 [0.808, 0.821]	held
089 P5h	LRU still leads FreeToken at gpt-oss 11%	sign	0.973 [0.962, 0.986]	failed
089 P5i	LRU still leads FreeToken at gpt-oss 25%	sign	1.045 [1.016, 1.068]	held
089 P6a	prefill at 2,048 tokens within $\pm 20\%$ of stock, gpt-oss	band	0.800	held (point)
089 P6b	prefill at 2,048 tokens within $\pm 20\%$ of stock, Qwen3	band	0.825	held (point)
089 P6c	decode at 2 and 4 sequences below stock's, gpt-oss	sign	0.778 / 0.778 / 0.773 / 0.784	held (point)
089 P6d	decode at 2 and 4 sequences below stock's, Qwen3	sign	0.779 / 0.789 / 0.775 / 0.792	held (point)
090 P1a	mean KL ≤ 0.01 nats, gpt-oss	threshold	0.0019	held
090 P1b	mean KL ≤ 0.01 nats, Qwen3	threshold	0.0005	held
090 P1c	99.9th-percentile KL ≤ 0.5 nats, gpt-oss	threshold	0.121	held
090 P1d	99.9th-percentile KL ≤ 0.5 nats, Qwen3	threshold	0.059	held
090 P2a	top-1 agreement $\geq 98\%$, gpt-oss	threshold	98.5	held
090 P2b	top-1 agreement $\geq 98\%$, Qwen3	threshold	99.3	held
090 P2c	stock's -n-cpu-moe agreement within 1 point of ours, gpt-oss	band	0.33	held
090 P2d	stock's -n-cpu-moe agreement within 1 point of ours, Qwen3	band	-0.07	held
090 P3a	mean NLL within $\pm 0.5\%$ of the reference, gpt-oss	band	0.16	held
090 P3b	mean NLL within $\pm 0.5\%$ of the reference, Qwen3	band	0.19	held
091 P1a	ours leads FreeToken, gpt-oss 11% (CI > 1)	sign	1.371 [1.346, 1.395]	held
091 P1b	ours leads FreeToken, gpt-oss 25% (CI > 1)	sign	1.296 [1.265, 1.326]	held
091 P1c	ours leads FreeToken, Qwen3 12.5% (CI > 1)	sign	1.033 [1.021, 1.045]	held
091 P1d	ours leads FreeToken, Qwen3 25% (CI > 1)	sign	1.045 [1.028, 1.063]	held
091 P1e	lead in 10-35%, gpt-oss 11%	band	37.1 [34.6, 39.5]	failed
091 P1f	lead in 10-35%, gpt-oss 25%	band	29.6 [26.5, 32.6]	held
091 P1g	lead in 0-15%, Qwen3 12.5%	band	3.3 [2.1, 4.5]	held
091 P1h	lead in 0-15%, Qwen3 25%	band	4.5 [2.8, 6.3]	held
091 P2a	ours $\geq 1.8 \times$ llama.cpp, gpt-oss 11%	threshold	1.810 [1.760, 1.860]	held (point)
091 P2b	ours $\geq 1.8 \times$ llama.cpp, gpt-oss 25%	threshold	2.620 [2.520, 2.740]	held
091 P2c	ours $\geq 1.8 \times$ llama.cpp, Qwen3 12.5%	threshold	1.780 [1.760, 1.810]	failed
091 P2d	ours $\geq 1.8 \times$ llama.cpp, Qwen3 25%	threshold	2.540 [2.490, 2.610]	held
091 P3a	the law's gpt-oss table fetches no more than the headline machine's	equality	0,0,1,2,2	failed
091 P3b	the law's Qwen3 table fetches no more than the headline machine's	equality	0,0,1,1,2,3,4,4,5	failed
091 P4a	ours at 25-45% of the limit, gpt-oss 11%	band	44.4 [43.3, 45.5]	held (point)
091 P4b	ours at 25-45% of the limit, gpt-oss 25%	band	29.9 [29.1, 30.8]	held
091 P4c	ours at 25-45% of the limit, Qwen3 12.5%	band	45.7 [45.1, 46.3]	failed
091 P4d	ours at 25-45% of the limit, Qwen3 25%	band	34.3 [33.5, 35.2]	held
091 P5	ours-first and FreeToken-first orders agree within 3%, every cell	band	1.6 / 2.0 / -0.6 / 0.1	held (point)
092 P1a	ours leads FreeToken, gpt-oss 11% (CI > 1)	sign	4.460 [4.250, 4.690]	held
092 P1b	ours leads FreeToken, Qwen3 12.5% (CI > 1)	sign	2.120 [2.020, 2.210]	held
092 P1c	ours leads FreeToken, Qwen3 25% (CI > 1)	sign	2.680 [2.530, 2.830]	held
092 P1d	lead in 10-35%, gpt-oss 11%	band	346 [325, 369]	failed
092 P1e	lead in 0-15%, Qwen3 12.5%	band	112 [102, 121]	failed
092 P1f	lead in 0-15%, Qwen3 25%	band	168 [153, 183]	failed

continued

Job	Clause	Type	Measured	Status
092 P2a	ours $\geq 1.8\times$ llama.cpp, gpt-oss 11%	threshold	1.640 [1.600, 1.670]	failed
092 P2b	ours $\geq 1.8\times$ llama.cpp, Qwen3 12.5%	threshold	1.560 [1.480, 1.630]	failed
092 P2c	ours $\geq 1.8\times$ llama.cpp, Qwen3 25%	threshold	2.180 [2.090, 2.270]	held
092 P3a	ours $\geq 1.3\times$ llama.cpp on Mixtral-8x7B, C = 2	threshold	1.015 [1.008, 1.024]	failed
092 P3b	ours $\geq 1.3\times$ llama.cpp on Mixtral-8x7B, C = 4	threshold	–	<i>untested</i>
092 P4a	the law’s gpt-oss table fetches no more than the headline machine’s	equality	0,0,0,0,0	held
092 P4b	the law’s Qwen3 table fetches no more than the headline machine’s	equality	0,0,0,0,0,0,0,0,0	held
092 P5a	ours-first and FreeToken-first orders agree within 3%, Qwen3 12.5%	band	7.0	failed
092 P5b	ours-first and FreeToken-first orders agree within 3%, Qwen3 25%	band	2.6	held (point)
093 P1a	oracle (W = all) faster than base, gpt-oss 11%	sign	0.923 [0.914, 0.933]	failed
093 P1b	oracle (W = all) faster than base, gpt-oss 25%	sign	1.398 [1.384, 1.412]	held
093 P1c	oracle (W = all) faster than base, Qwen3 12.5%	sign	0.916 [0.911, 0.922]	failed
093 P1d	oracle (W = all) faster than base, Qwen3 25%	sign	1.247 [1.235, 1.259]	held
093 P2a	oracle gain 20-80%, gpt-oss 11%	band	-7.7 [-8.6, -6.7]	failed
093 P2b	oracle gain 20-80%, gpt-oss 25%	band	39.8 [38.4, 41.2]	held
093 P2c	oracle gain 20-80%, Qwen3 12.5%	band	-8.4 [-8.9, -7.8]	failed
093 P2d	oracle gain 20-80%, Qwen3 25%	band	24.7 [23.5, 25.9]	held
093 P3a	W = 16 captures $\geq$ half of the W = 0 gain, gpt-oss 11%	threshold	–	<i>untested</i>
093 P3b	W = 16 captures $\geq$ half of the W = 0 gain, gpt-oss 25%	threshold	87	held (point)
093 P3c	W = 16 captures $\geq$ half of the W = 0 gain, Qwen3 12.5%	threshold	–	<i>untested</i>
093 P3d	W = 16 captures $\geq$ half of the W = 0 gain, Qwen3 25%	threshold	100	held (point)
093 P3e	W = 4 captures $\geq$ half of the W = 0 gain, gpt-oss 11%	threshold	–	<i>untested</i>
093 P3f	W = 4 captures $\geq$ half of the W = 0 gain, Qwen3 12.5%	threshold	–	<i>untested</i>
093 P4a	ours with the oracle at 45-65% of the limit, gpt-oss 11%	band	44.0 [43.2, 44.7]	failed
093 P4b	ours with the oracle at 45-65% of the limit, gpt-oss 25%	band	43.7 [42.5, 44.8]	failed
093 P4c	ours with the oracle at 45-65% of the limit, Qwen3 12.5%	band	44.8 [44.1, 45.5]	failed
093 P4d	ours with the oracle at 45-65% of the limit, Qwen3 25%	band	44.5 [43.3, 45.8]	failed
093 P5a	oracle hit rate within 3 points of the optimum’s, gpt-oss 11%	band	-5.9	failed
093 P5b	oracle hit rate within 3 points of the optimum’s, gpt-oss 25%	band	3.1	failed
093 P5c	oracle hit rate within 3 points of the optimum’s, Qwen3 12.5%	band	-9.2	failed
093 P5d	oracle hit rate within 3 points of the optimum’s, Qwen3 25%	band	0.4	held
093 P5e	oracle admissions within 30% of the optimum’s reads, gpt-oss 11%	band	-9	held
093 P5f	oracle admissions within 30% of the optimum’s reads, gpt-oss 25%	band	31	failed
093 P5g	oracle admissions within 30% of the optimum’s reads, Qwen3 12.5%	band	-35	failed
093 P5h	oracle admissions within 30% of the optimum’s reads, Qwen3 25%	band	7	held
093 P6a	no-overlap 5-25% slower than base, gpt-oss 11%	band	-0.5 [-1.0, -0.1]	failed
093 P6b	no-overlap 5-25% slower than base, gpt-oss 25%	band	0.2 [0.1, 0.2]	failed
093 P6c	no-overlap 5-25% slower than base, Qwen3 12.5%	band	-0.0 [-0.1, 0.0]	failed
093 P6d	no-overlap 5-25% slower than base, Qwen3 25%	band	0.1 [0.0, 0.2]	failed
093 P6e	all-CPU 0-10% slower than base, gpt-oss 11%	band	7.8 [7.2, 8.4]	held
093 P6f	all-CPU 0-10% slower than base, gpt-oss 25%	band	9.3 [8.7, 9.8]	held
093 P6g	all-CPU 0-10% slower than base, Qwen3 12.5%	band	14.7 [14.1, 15.3]	failed
093 P6h	all-CPU 0-10% slower than base, Qwen3 25%	band	14.6 [14.0, 15.2]	failed
093 P7a	oracle gain below 15% at the GPU-bound cell, gpt-oss 40%	threshold	48.5 [46.2, 50.9]	failed
093 P7b	oracle gain below 15% at the GPU-bound cell, Qwen3 43.75%	threshold	56.5 [53.3, 59.2]	failed

continued

Job	Clause	Type	Measured	Status
094 P1a	bypass oracle's reads within 15% of the optimum's, gpt-oss 11%	band	30.8	failed
094 P1b	bypass oracle's reads within 15% of the optimum's, gpt-oss 25%	band	31.1	failed
094 P1c	bypass oracle's reads within 15% of the optimum's, gpt-oss 40%	band	29.0	failed
094 P1d	bypass oracle's reads within 15% of the optimum's, Qwen3 12.5%	band	43.4	failed
094 P1e	bypass oracle's reads within 15% of the optimum's, Qwen3 25%	band	46.6	failed
094 P1f	bypass oracle's reads within 15% of the optimum's, Qwen3 43.75%	band	38.5	failed
094 P2a	bypass oracle faster than base, gpt-oss 11%	sign	1.033 [1.029, 1.038]	held
094 P2b	bypass oracle faster than base, gpt-oss 25%	sign	1.181 [1.175, 1.187]	held
094 P2c	bypass oracle faster than base, gpt-oss 40%	sign	1.232 [1.220, 1.243]	held
094 P2d	bypass oracle faster than base, Qwen3 12.5%	sign	1.002 [0.997, 1.006]	held (point)
094 P2e	bypass oracle faster than base, Qwen3 25%	sign	1.131 [1.122, 1.139]	held
094 P2f	bypass oracle faster than base, Qwen3 43.75%	sign	1.189 [1.178, 1.200]	held
094 P2g	bypass gain 10-45%, gpt-oss 11%	band	3.3 [2.9, 3.8]	failed
094 P2h	bypass gain 10-45%, gpt-oss 25%	band	18.1 [17.5, 18.7]	held
094 P2i	bypass gain 10-45%, Qwen3 12.5%	band	0.2 [-0.3, 0.6]	failed
094 P2j	bypass gain 10-45%, Qwen3 25%	band	13.1 [12.2, 13.9]	held
094 P3a	bypass hit rate within 4 points of the optimum's, gpt-oss 11%	band	-8.2	failed
094 P3b	bypass hit rate within 4 points of the optimum's, gpt-oss 25%	band	-3.2	held
094 P3c	bypass hit rate within 4 points of the optimum's, gpt-oss 40%	band	-1.4	held
094 P3d	bypass hit rate within 4 points of the optimum's, Qwen3 12.5%	band	-10.9	failed
094 P3e	bypass hit rate within 4 points of the optimum's, Qwen3 25%	band	-5.3	failed
094 P3f	bypass hit rate within 4 points of the optimum's, Qwen3 43.75%	band	-1.3	held
094 P4a	law over-predicts the bypass oracle by 10-40%, gpt-oss 11%	band	6.9	failed
094 P4b	law over-predicts the bypass oracle by 10-40%, gpt-oss 25%	band	5.8	failed
094 P4c	law over-predicts the bypass oracle by 10-40%, Qwen3 12.5%	band	4.7	failed
094 P4d	law over-predicts the bypass oracle by 10-40%, Qwen3 25%	band	8.5	failed
094 P5a	bypass beats prefetch by $\geq 15\%$, gpt-oss 11%	threshold	6.9	failed
094 P5b	bypass beats prefetch by $\geq 15\%$, Qwen3 12.5%	threshold	5.9	failed
094 P5c	bypass and prefetch within 15%, gpt-oss 25%	band	-27.6	failed
094 P5d	bypass and prefetch within 15%, gpt-oss 40%	band	-21.3	failed
094 P5e	bypass and prefetch within 15%, Qwen3 25%	band	-19.7	failed
094 P5f	bypass and prefetch within 15%, Qwen3 43.75%	band	-26.6	failed
094 P6a	W = 16 keeps $\geq 70\%$ of the W = all gain, gpt-oss 11%	threshold	176 [165, 190]	held
094 P6b	W = 16 keeps $\geq 70\%$ of the W = all gain, gpt-oss 25%	threshold	63	failed
094 P6c	W = 16 keeps $\geq 70\%$ of the W = all gain, gpt-oss 40%	threshold	34	failed
094 P6d	W = 16 keeps $\geq 70\%$ of the W = all gain, Qwen3 12.5%	threshold	–	<i>untested</i>
094 P6e	W = 16 keeps $\geq 70\%$ of the W = all gain, Qwen3 25%	threshold	94 [91, 96]	held
094 P6f	W = 16 keeps $\geq 70\%$ of the W = all gain, Qwen3 43.75%	threshold	53	failed
095 P1a	fetch oracle's reads within 10% of the simulation, gpt-oss 11%	band	-0.2	held
095 P1b	fetch oracle's reads within 10% of the simulation, gpt-oss 25%	band	-6.2	held
095 P1c	fetch oracle's reads within 10% of the simulation, gpt-oss 40%	band	-17.2	failed
095 P1d	fetch oracle's reads within 10% of the simulation, Qwen3 12.5%	band	-0.5	held

continued

Job	Clause	Type	Measured	Status
095 P1e	fetch oracle's reads within 10% of the simulation, Qwen3 25%	band	-4.4	held
095 P1f	fetch oracle's reads within 10% of the simulation, Qwen3 43.75%	band	-16.9	failed
095 P1g	fetch oracle's hit rate within 3 points of the simulation, gpt-oss 11%	band	0.1	held
095 P1h	fetch oracle's hit rate within 3 points of the simulation, gpt-oss 25%	band	0.8	held
095 P1i	fetch oracle's hit rate within 3 points of the simulation, gpt-oss 40%	band	1.1	held
095 P1j	fetch oracle's hit rate within 3 points of the simulation, Qwen3 12.5%	band	0.1	held
095 P1k	fetch oracle's hit rate within 3 points of the simulation, Qwen3 25%	band	0.5	held
095 P1l	fetch oracle's hit rate within 3 points of the simulation, Qwen3 43.75%	band	0.8	held
095 P2a	fetch oracle faster than base, gpt-oss 11%	sign	1.286 [1.279, 1.294]	held
095 P2b	fetch oracle faster than base, gpt-oss 25%	sign	1.316 [1.307, 1.326]	held
095 P2c	fetch oracle faster than base, gpt-oss 40%	sign	1.254 [1.242, 1.265]	held
095 P2d	fetch oracle faster than base, Qwen3 12.5%	sign	1.345 [1.338, 1.351]	held
095 P2e	fetch oracle faster than base, Qwen3 25%	sign	1.379 [1.368, 1.390]	held
095 P2f	fetch oracle faster than base, Qwen3 43.75%	sign	1.264 [1.251, 1.276]	held
095 P2g	law within 10% on the fetch oracle, gpt-oss 11%	band	-7.6 [-9.0, -6.3]	held
095 P2h	law within 10% on the fetch oracle, gpt-oss 25%	band	-13.4	failed
095 P2i	law within 10% on the fetch oracle, gpt-oss 40%	band	-13.6	failed
095 P2j	law within 10% on the fetch oracle, Qwen3 12.5%	band	-8.9 [-10.0, -7.8]	held
095 P2k	law within 10% on the fetch oracle, Qwen3 25%	band	-14.9	failed
095 P2l	law within 10% on the fetch oracle, Qwen3 43.75%	band	-16.9	failed
095 P3a	both2's reads within 25% of the simulation, gpt-oss 11%	band	-3.7	held
095 P3b	both2's reads within 25% of the simulation, gpt-oss 25%	band	-3.9	held
095 P3c	both2's reads within 25% of the simulation, gpt-oss 40%	band	-12.3	held
095 P3d	both2's reads within 25% of the simulation, Qwen3 12.5%	band	-1.9	held
095 P3e	both2's reads within 25% of the simulation, Qwen3 25%	band	-4.9	held
095 P3f	both2's reads within 25% of the simulation, Qwen3 43.75%	band	-8.7	held
095 P3g	both2's hit rate within 5 points of the simulation, gpt-oss 11%	band	-0.7	held
095 P3h	both2's hit rate within 5 points of the simulation, gpt-oss 25%	band	-1.6	held
095 P3i	both2's hit rate within 5 points of the simulation, gpt-oss 40%	band	-0.5	held
095 P3j	both2's hit rate within 5 points of the simulation, Qwen3 12.5%	band	0.2	held
095 P3k	both2's hit rate within 5 points of the simulation, Qwen3 25%	band	-0.4	held
095 P3l	both2's hit rate within 5 points of the simulation, Qwen3 43.75%	band	-0.3	held
095 P4a	both2 at 55-85% of the limit, gpt-oss 11%	band	63.9 [62.8, 65.0]	held
095 P4b	both2 at 55-85% of the limit, gpt-oss 25%	band	44.4	failed
095 P4c	both2 at 55-85% of the limit, Qwen3 12.5%	band	68.0 [67.1, 69.0]	held
095 P4d	both2 at 55-85% of the limit, Qwen3 25%	band	50.9	failed
095 P4e	both2 faster than the hit-optimal oracle, gpt-oss 11%	sign	1.532 [1.519, 1.546]	held
095 P4f	both2 faster than the hit-optimal oracle, gpt-oss 25%	sign	1.310 [1.291, 1.329]	held
095 P4g	both2 faster than the hit-optimal oracle, Qwen3 12.5%	sign	2.045 [2.016, 2.075]	held
095 P4h	both2 faster than the hit-optimal oracle, Qwen3 25%	sign	1.388 [1.374, 1.402]	held
095 P5a	law over-predicts both2 by 15-50%, gpt-oss 11%	band	12.7	failed
095 P5b	law over-predicts both2 by 15-50%, gpt-oss 25%	band	25.7 [23.2, 28.0]	held
095 P5c	law over-predicts both2 by 15-50%, Qwen3 12.5%	band	7.1	failed
095 P5d	law over-predicts both2 by 15-50%, Qwen3 25%	band	20.8 [17.8, 24.1]	held

continued

Job	Clause	Type	Measured	Status
095 P6a	both2 gains $\geq 25\%$ over base, gpt-oss 11%	threshold	39.4 [38.6, 40.3]	held
095 P6b	both2 gains $\geq 25\%$ over base, Qwen3 12.5%	threshold	42.1 [41.3, 42.9]	held
095 P7a	paced both3p within 10% of both2, gpt-oss 11%	band	4.2 [3.7, 4.7]	held
095 P7b	paced both3p within 10% of both2, gpt-oss 25%	band	11.8	failed
095 P7c	paced both3p within 10% of both2, gpt-oss 40%	band	7.3 [6.8, 7.8]	held
095 P7d	paced both3p within 10% of both2, Qwen3 12.5%	band	-0.5 [-0.8, -0.2]	held
095 P7e	paced both3p within 10% of both2, Qwen3 25%	band	13.0	failed
095 P7f	paced both3p within 10% of both2, Qwen3 43.75%	band	6.8 [6.1, 7.5]	held
096O4 P1a-gpt-oss11	fetch reads within 6% of the simulation, gpt-oss 11% (O4)	band	1.5	held (point)
096O4 P1c-gpt-oss11	fetch hit rate within 2 points of the simulation, gpt-oss 11% (O4)	band	-0.4	held (point)
096O4 P1b-gpt-oss11	both2 reads within 15% of the simulation, gpt-oss 11% (O4)	band	1.6	held (point)
096O4 P3a-gpt-oss11	foa reads 3-22% fewer than base, gpt-oss 11% (O4)	band	6.4	held (point)
096O4 P2-gpt-oss11	fetch $1.15\text{-}1.45\times$ base, gpt-oss 11% (O4)	band	1.361 [1.353, 1.369]	held
096O4 P3b-gpt-oss11	foa $0.97\text{-}1.12\times$ base, gpt-oss 11% (O4)	band	1.020 [1.019, 1.022]	held
096O4 P4-gpt-oss11	best prefetch $>$ fetch $>$ foa, gpt-oss 11% (O4)	sign	0.122	held
096O4 P7b-gpt-oss11	lead2 faster than bypass, gpt-oss 11% (O4)	sign	1.300	held (point)
096O4 P5a-gpt-oss11	nb2 reads $1.25\text{-}1.9\times$ lead2, gpt-oss 11% (O4)	band	1.25	failed
096O4 P5b-gpt-oss11	nb2 slower than lead2, gpt-oss 11% (O4)	sign	0.788	held (point)
096O4 P5c-gpt-oss11	nb2 at most $1.05\times$ base, gpt-oss 11% (O4)	threshold	1.049 [1.036, 1.062]	held (point)
096O4 P7a-gpt-oss11	bypass gains at most 8%, gpt-oss 11% (O4)	threshold	1.024 [1.019, 1.028]	held
096O4 P6b-gpt-oss11	nb2 within 10% of hitopt, gpt-oss 11% (O4)	band	0.7	held (point)
096O4 P9a-gpt-oss11	bytes share 25-55% of the gap, gpt-oss 11% (O4)	band	51.6	held (point)
096O4 P9b-gpt-oss11	overlap share 5-30% of the gap, gpt-oss 11% (O4)	band	12.8	held (point)
096O4 P1a-gpt-oss25	fetch reads within 6% of the simulation, gpt-oss 25% (O4)	band	3.1	held (point)
096O4 P1c-gpt-oss25	fetch hit rate within 2 points of the simulation, gpt-oss 25% (O4)	band	-0.4	held (point)
096O4 P1b-gpt-oss25	both2 reads within 15% of the simulation, gpt-oss 25% (O4)	band	-0.3	held (point)
096O4 P3a-gpt-oss25	foa reads 3-22% fewer than base, gpt-oss 25% (O4)	band	13.4	held (point)
096O4 P2-gpt-oss25	fetch $1.15\text{-}1.45\times$ base, gpt-oss 25% (O4)	band	1.426 [1.414, 1.438]	held
096O4 P3b-gpt-oss25	foa $0.97\text{-}1.12\times$ base, gpt-oss 25% (O4)	band	1.030 [1.029, 1.032]	held

continued

Job	Clause	Type	Measured	Status
096O4 P4-gpt-oss25	best prefetch > fetch > foa, gpt-oss 25% (O4)	sign	0.360	held
096O4 P7b-gpt-oss25	lead2 faster than bypass, gpt-oss 25% (O4)	sign	1.353	held (point)
096O4 P5a-gpt-oss25	nb2 reads 1.25-1.9× lead2, gpt-oss 25% (O4)	band	1.36	held (point)
096O4 P6b-gpt-oss25	nb2 within 10% of hitopt, gpt-oss 25% (O4)	band	1.6	held (point)
096O4 P8-gpt-oss25	hitoptp faster than hitopt, gpt-oss 25% (O4)	sign	1.200	held (point)
096O4 P9a-gpt-oss25	bytes share 25-55% of the gap, gpt-oss 25% (O4)	band	44.1	held (point)
096O4 P9b-gpt-oss25	overlap share 5-30% of the gap, gpt-oss 25% (O4)	band	21.9	held (point)
096O4 P1a-gpt-oss40	fetch reads within 6% of the simulation, gpt-oss 40% (O4)	band	2.9	held (point)
096O4 P1c-gpt-oss40	fetch hit rate within 2 points of the simulation, gpt-oss 40% (O4)	band	-0.2	held (point)
096O4 P1b-gpt-oss40	both2 reads within 15% of the simulation, gpt-oss 40% (O4)	band	-9.8	held (point)
096O4 P3a-gpt-oss40	foa reads 3-22% fewer than base, gpt-oss 40% (O4)	band	21.0	held (point)
096O4 P5a-gpt-oss40	nb2 reads 1.25-1.9× lead2, gpt-oss 40% (O4)	band	1.29	held (point)
096O4 P6b-gpt-oss40	nb2 within 10% of hitopt, gpt-oss 40% (O4)	band	2.0	held (point)
096O4 P1a-Qwen312.5	fetch reads within 6% of the simulation, Qwen3 12.5% (O4)	band	0.4	held (point)
096O4 P1c-Qwen312.5	fetch hit rate within 2 points of the simulation, Qwen3 12.5% (O4)	band	-0.2	held (point)
096O4 P1b-Qwen312.5	both2 reads within 15% of the simulation, Qwen3 12.5% (O4)	band	0.3	held (point)
096O4 P3a-Qwen312.5	foa reads 3-22% fewer than base, Qwen3 12.5% (O4)	band	4.0	held (point)
096O4 P2-Qwen312.5	fetch 1.15-1.45× base, Qwen3 12.5% (O4)	band	1.438 [1.430, 1.445]	held
096O4 P3b-Qwen312.5	foa 0.97-1.12× base, Qwen3 12.5% (O4)	band	1.015 [1.014, 1.016]	held
096O4 P4-Qwen312.5	best prefetch > fetch > foa, Qwen3 12.5% (O4)	sign	0.067	held
096O4 P7b-Qwen312.5	lead2 faster than bypass, Qwen3 12.5% (O4)	sign	1.252	held (point)
096O4 P5a-Qwen312.5	nb2 reads 1.25-1.9× lead2, Qwen3 12.5% (O4)	band	1.09	failed
096O4 P5b-Qwen312.5	nb2 slower than lead2, Qwen3 12.5% (O4)	sign	0.923	held (point)
096O4 P5c-Qwen312.5	nb2 at most 1.05× base, Qwen3 12.5% (O4)	threshold	1.149 [1.140, 1.158]	failed

continued

Job	Clause	Type	Measured	Status
096O4 P7a- Qwen312.5	bypass gains at most 8%, Qwen3 12.5% (O4)	threshold	0.995 [0.992, 0.998]	held
096O4 P6a- Qwen312.5	nb2 faster than hitopt, Qwen3 12.5% (O4)	sign	1.542	held (point)
096O4 P9a- Qwen312.5	bytes share 25-55% of the gap, Qwen3 12.5% (O4)	band	60.2	failed
096O4 P9b- Qwen312.5	overlap share 5-30% of the gap, Qwen3 12.5% (O4)	band	6.9	held (point)
096O4 P1a- Qwen325	fetch reads within 6% of the simulation, Qwen3 25% (O4)	band	0.7	held (point)
096O4 P1c- Qwen325	fetch hit rate within 2 points of the simulation, Qwen3 25% (O4)	band	-0.1	held (point)
096O4 P1b- Qwen325	both2 reads within 15% of the simulation, Qwen3 25% (O4)	band	0.2	held (point)
096O4 P3a- Qwen325	foa reads 3-22% fewer than base, Qwen3 25% (O4)	band	6.9	held (point)
096O4 P2- Qwen325	fetch 1.15-1.45 $\times$ base, Qwen3 25% (O4)	band	1.513 [1.500, 1.525]	failed
096O4 P3b- Qwen325	foa 0.97-1.12 $\times$ base, Qwen3 25% (O4)	band	1.016 [1.015, 1.017]	held
096O4 P4- Qwen325	best prefetch > fetch > foa, Qwen3 25% (O4)	sign	0.271	held
096O4 P7b- Qwen325	lead2 faster than bypass, Qwen3 25% (O4)	sign	1.410	held (point)
096O4 P5a- Qwen325	nb2 reads 1.25-1.9 $\times$ lead2, Qwen3 25% (O4)	band	1.28	held (point)
096O4 P8- Qwen325	hitoptp faster than hitopt, Qwen3 25% (O4)	sign	1.048	held (point)
096O4 P9a- Qwen325	bytes share 25-55% of the gap, Qwen3 25% (O4)	band	53.4	held (point)
096O4 P9b- Qwen325	overlap share 5-30% of the gap, Qwen3 25% (O4)	band	16.8	held (point)
096O4 P1a- Qwen343.75	fetch reads within 6% of the simulation, Qwen3 43.75% (O4)	band	1.4	held (point)
096O4 P1c- Qwen343.75	fetch hit rate within 2 points of the simulation, Qwen3 43.75% (O4)	band	-0.0	held (point)
096O4 P1b- Qwen343.75	both2 reads within 15% of the simulation, Qwen3 43.75% (O4)	band	-4.6	held (point)
096O4 P3a- Qwen343.75	foa reads 3-22% fewer than base, Qwen3 43.75% (O4)	band	16.3	held (point)
096O4 P5a- Qwen343.75	nb2 reads 1.25-1.9 $\times$ lead2, Qwen3 43.75% (O4)	band	1.31	held (point)
096O5 P1a-gpt- oss11	fetch reads within 6% of the simulation, gpt-oss 11% (O5)	band	1.5	held (point)
096O5 P1c-gpt- oss11	fetch hit rate within 2 points of the simulation, gpt-oss 11% (O5)	band	-0.4	held (point)
096O5 P1b-gpt- oss11	both2 reads within 15% of the simulation, gpt-oss 11% (O5)	band	1.6	held (point)

continued

Job	Clause	Type	Measured	Status
096O5 P3a-gpt-oss11	foa reads 3-22% fewer than base, gpt-oss 11% (O5)	band	6.6	held (point)
096O5 P2-gpt-oss11	fetch 1.15-1.45 $\times$ base, gpt-oss 11% (O5)	band	1.159 [1.154, 1.165]	held
096O5 P3b-gpt-oss11	foa 0.97-1.12 $\times$ base, gpt-oss 11% (O5)	band	0.998 [0.994, 1.001]	held
096O5 P4-gpt-oss11	best prefetch > fetch > foa, gpt-oss 11% (O5)	sign	0.130	held
096O5 P7b-gpt-oss11	lead2 faster than bypass, gpt-oss 11% (O5)	sign	1.042	held (point)
096O5 P5a-gpt-oss11	nb2 reads 1.25-1.9 $\times$ lead2, gpt-oss 11% (O5)	band	1.27	held (point)
096O5 P5b-gpt-oss11	nb2 slower than lead2, gpt-oss 11% (O5)	sign	0.624	held (point)
096O5 P5c-gpt-oss11	nb2 at most 1.05 $\times$ base, gpt-oss 11% (O5)	threshold	0.644 [0.635, 0.652]	held
096O5 P7a-gpt-oss11	bypass gains at most 8%, gpt-oss 11% (O5)	threshold	0.990 [0.986, 0.994]	held
096O5 P6b-gpt-oss11	nb2 within 10% of hitopt, gpt-oss 11% (O5)	band	0.9	held (point)
096O5 P9a-gpt-oss11	bytes share 25-55% of the gap, gpt-oss 11% (O5)	band	25.8	held (point)
096O5 P9b-gpt-oss11	overlap share 5-30% of the gap, gpt-oss 11% (O5)	band	17.2	held (point)
096O5 P1a-gpt-oss25	fetch reads within 6% of the simulation, gpt-oss 25% (O5)	band	3.1	held (point)
096O5 P1c-gpt-oss25	fetch hit rate within 2 points of the simulation, gpt-oss 25% (O5)	band	-0.4	held (point)
096O5 P1b-gpt-oss25	both2 reads within 15% of the simulation, gpt-oss 25% (O5)	band	-0.3	held (point)
096O5 P3a-gpt-oss25	foa reads 3-22% fewer than base, gpt-oss 25% (O5)	band	15.3	held (point)
096O5 P2-gpt-oss25	fetch 1.15-1.45 $\times$ base, gpt-oss 25% (O5)	band	1.227 [1.220, 1.234]	held
096O5 P3b-gpt-oss25	foa 0.97-1.12 $\times$ base, gpt-oss 25% (O5)	band	0.974 [0.971, 0.978]	held
096O5 P4-gpt-oss25	best prefetch > fetch > foa, gpt-oss 25% (O5)	sign	0.245	held
096O5 P7b-gpt-oss25	lead2 faster than bypass, gpt-oss 25% (O5)	sign	1.020	held (point)
096O5 P5a-gpt-oss25	nb2 reads 1.25-1.9 $\times$ lead2, gpt-oss 25% (O5)	band	1.36	held (point)
096O5 P6b-gpt-oss25	nb2 within 10% of hitopt, gpt-oss 25% (O5)	band	1.7	held (point)
096O5 P8-gpt-oss25	hitopt faster than hitopt, gpt-oss 25% (O5)	sign	1.420	held (point)
096O5 P9a-gpt-oss25	bytes share 25-55% of the gap, gpt-oss 25% (O5)	band	26.6	held (point)

continued

Job	Clause	Type	Measured	Status
096O5 P9b-gpt-oss25	overlap share 5-30% of the gap, gpt-oss 25% (O5)	band	21.3	held (point)
096O5 P1a-gpt-oss40	fetch reads within 6% of the simulation, gpt-oss 40% (O5)	band	2.9	held (point)
096O5 P1c-gpt-oss40	fetch hit rate within 2 points of the simulation, gpt-oss 40% (O5)	band	-0.2	held (point)
096O5 P1b-gpt-oss40	both2 reads within 15% of the simulation, gpt-oss 40% (O5)	band	-9.8	held (point)
096O5 P3a-gpt-oss40	foa reads 3-22% fewer than base, gpt-oss 40% (O5)	band	22.3	failed
096O5 P5a-gpt-oss40	nb2 reads 1.25-1.9 $\times$ lead2, gpt-oss 40% (O5)	band	1.29	held (point)
096O5 P6b-gpt-oss40	nb2 within 10% of hitopt, gpt-oss 40% (O5)	band	2.5	held (point)
096O5 P1a-Qwen312.5	fetch reads within 6% of the simulation, Qwen3 12.5% (O5)	band	0.4	held (point)
096O5 P1c-Qwen312.5	fetch hit rate within 2 points of the simulation, Qwen3 12.5% (O5)	band	-0.2	held (point)
096O5 P1b-Qwen312.5	both2 reads within 15% of the simulation, Qwen3 12.5% (O5)	band	0.3	held (point)
096O5 P3a-Qwen312.5	foa reads 3-22% fewer than base, Qwen3 12.5% (O5)	band	4.6	held (point)
096O5 P2-Qwen312.5	fetch 1.15-1.45 $\times$ base, Qwen3 12.5% (O5)	band	1.199 [1.193, 1.205]	held
096O5 P3b-Qwen312.5	foa 0.97-1.12 $\times$ base, Qwen3 12.5% (O5)	band	1.006 [1.004, 1.007]	held
096O5 P4-Qwen312.5	best prefetch > fetch > foa, Qwen3 12.5% (O5)	sign	0.048	held
096O5 P7b-Qwen312.5	lead2 faster than bypass, Qwen3 12.5% (O5)	sign	1.084	held (point)
096O5 P5a-Qwen312.5	nb2 reads 1.25-1.9 $\times$ lead2, Qwen3 12.5% (O5)	band	1.14	failed
096O5 P5b-Qwen312.5	nb2 slower than lead2, Qwen3 12.5% (O5)	sign	0.753	held (point)
096O5 P5c-Qwen312.5	nb2 at most 1.05 $\times$ base, Qwen3 12.5% (O5)	threshold	0.766 [0.760, 0.773]	held
096O5 P7a-Qwen312.5	bypass gains at most 8%, Qwen3 12.5% (O5)	threshold	0.938 [0.934, 0.943]	held
096O5 P6a-Qwen312.5	nb2 faster than hitopt, Qwen3 12.5% (O5)	sign	1.413	held (point)
096O5 P9a-Qwen312.5	bytes share 25-55% of the gap, Qwen3 12.5% (O5)	band	31.6	held (point)
096O5 P9b-Qwen312.5	overlap share 5-30% of the gap, Qwen3 12.5% (O5)	band	6.9	held (point)
096O5 P1a-Qwen325	fetch reads within 6% of the simulation, Qwen3 25% (O5)	band	0.7	held (point)

continued

Job	Clause	Type	Measured	Status
096O5 P1c- Qwen325	fetch hit rate within 2 points of the simulation, Qwen3 25% (O5)	band	-0.1	held (point)
096O5 P1b- Qwen325	both2 reads within 15% of the simulation, Qwen3 25% (O5)	band	0.2	held (point)
096O5 P3a- Qwen325	foa reads 3-22% fewer than base, Qwen3 25% (O5)	band	8.4	held (point)
096O5 P2- Qwen325	fetch $1.15\text{--}1.45\times$ base, Qwen3 25% (O5)	band	1.256 [1.247, 1.265]	held
096O5 P3b- Qwen325	foa $0.97\text{--}1.12\times$ base, Qwen3 25% (O5)	band	0.999 [0.997, 1.001]	held
096O5 P4- Qwen325	best prefetch > fetch > foa, Qwen3 25% (O5)	sign	0.182	held
096O5 P7b- Qwen325	lead2 faster than bypass, Qwen3 25% (O5)	sign	1.004	held (point)
096O5 P5a- Qwen325	nb2 reads $1.25\text{--}1.9\times$ lead2, Qwen3 25% (O5)	band	1.30	held (point)
096O5 P8- Qwen325	hitoptp faster than hitopt, Qwen3 25% (O5)	sign	1.415	held (point)
096O5 P9a- Qwen325	bytes share 25-55% of the gap, Qwen3 25% (O5)	band	31.1	held (point)
096O5 P9b- Qwen325	overlap share 5-30% of the gap, Qwen3 25% (O5)	band	16.3	held (point)
096O5 P1a- Qwen343.75	fetch reads within 6% of the simulation, Qwen3 43.75% (O5)	band	1.4	held (point)
096O5 P1c- Qwen343.75	fetch hit rate within 2 points of the simulation, Qwen3 43.75% (O5)	band	-0.0	held (point)
096O5 P1b- Qwen343.75	both2 reads within 15% of the simulation, Qwen3 43.75% (O5)	band	-4.6	held (point)
096O5 P3a- Qwen343.75	foa reads 3-22% fewer than base, Qwen3 43.75% (O5)	band	18.8	held (point)
096O5 P5a- Qwen343.75	nb2 reads $1.25\text{--}1.9\times$ lead2, Qwen3 43.75% (O5)	band	1.32	held (point)
096 P10- gpt-oss11	fetch/base differs by at most 0.15 between hosts, gpt-oss 11%	threshold	0.201	failed
096 P10- gpt-oss25	fetch/base differs by at most 0.15 between hosts, gpt-oss 25%	threshold	0.199	failed
096 P10- gpt-oss40	fetch/base differs by at most 0.15 between hosts, gpt-oss 40%	threshold	0.135	held (point)
096 P10- Qwen312.5	fetch/base differs by at most 0.15 between hosts, Qwen3 12.5%	threshold	0.239	failed
096 P10- Qwen325	fetch/base differs by at most 0.15 between hosts, Qwen3 25%	threshold	0.257	failed
096 P10- Qwen343.75	fetch/base differs by at most 0.15 between hosts, Qwen3 43.75%	threshold	0.152	failed
098 P1- gpt-oss11	FreeToken's best at most $1.10\times$ its Table 1 setting, gpt-oss 11%	threshold	1.042 [1.038, 1.049]	held
098 P2- gpt-oss11	ours leads FreeToken's best by at least 5%, gpt-oss 11%	threshold	1.173 [1.157, 1.187]	held
098 P3- gpt-oss11- cap1	fetch cap 1 within 8% of the calibrated split, gpt-oss 11%	band	-0.9	held (point)
098 P3- gpt-oss11- cap2	fetch cap 2 within 8% of the calibrated split, gpt-oss 11%	band	-5.2	held (point)
098 P4- gpt-oss11	8 threads slower than default or within 3%, gpt-oss 11%	threshold	4.2	failed

continued

Job	Clause	Type	Measured	Status
098 P5-gpt-oss11	FreeToken’s best backend is hybrid, gpt-oss 11%	equality	1	held
098 P6-gpt-oss11	FreeToken 27-30 GiB, ours below it, gpt-oss 11%	band	28.4	held
098 P1-gpt-oss25	FreeToken’s best at most 1.10× its Table 1 setting, gpt-oss 25%	threshold	1.030 [1.016, 1.045]	held
098 P2-gpt-oss25	ours leads FreeToken’s best by at least 5%, gpt-oss 25%	threshold	1.167 [1.147, 1.189]	held
098 P3-gpt-oss25-cap1	fetch cap 1 within 8% of the calibrated split, gpt-oss 25%	band	-0.3	held (point)
098 P3-gpt-oss25-cap2	fetch cap 2 within 8% of the calibrated split, gpt-oss 25%	band	-3.5	held (point)
098 P4-gpt-oss25	8 threads slower than default or within 3%, gpt-oss 25%	threshold	1.6	held (point)
098 P5-gpt-oss25	FreeToken’s best backend is hybrid, gpt-oss 25%	equality	0	failed
098 P6-gpt-oss25	FreeToken 27-30 GiB, ours below it, gpt-oss 25%	band	28.4	held
098 P1-gpt-oss40	FreeToken’s best at most 1.10× its Table 1 setting, gpt-oss 40%	threshold	1.000 [1.000, 1.000]	held
098 P2-gpt-oss40	ours / FreeToken-best in 0.95-1.15, gpt-oss 40%	band	1.094 [1.078, 1.112]	held
098 P3-gpt-oss40-cap1	fetch cap 1 within 8% of the calibrated split, gpt-oss 40%	band	0.3	held (point)
098 P3-gpt-oss40-cap2	fetch cap 2 within 8% of the calibrated split, gpt-oss 40%	band	-0.8	held (point)
098 P4-gpt-oss40	8 threads slower than default or within 3%, gpt-oss 40%	threshold	0.2	held (point)
098 P5-gpt-oss40	FreeToken’s best backend is offload, gpt-oss 40%	equality	1	held
098 P6-gpt-oss40	FreeToken 27-30 GiB, ours below it, gpt-oss 40%	band	28.4	held
098 P1-Qwen312.5	FreeToken’s best at most 1.10× its Table 1 setting, Qwen3 12.5%	threshold	1.011 [1.009, 1.014]	held
098 P2-Qwen312.5	ours / FreeToken-best in 0.90-1.12, Qwen3 12.5%	band	1.020 [1.012, 1.028]	held
098 P3-Qwen312.5-cap1	fetch cap 1 within 8% of the calibrated split, Qwen3 12.5%	band	-9.4	failed
098 P3-Qwen312.5-cap2	fetch cap 2 within 8% of the calibrated split, Qwen3 12.5%	band	-6.5	held (point)
098 P4-Qwen312.5	8 threads slower than default or within 3%, Qwen3 12.5%	threshold	1.1	held (point)
098 P5-Qwen312.5	FreeToken’s best backend is hybrid, Qwen3 12.5%	equality	1	held
098 P6-Qwen312.5	FreeToken 27-30 GiB, ours below it, Qwen3 12.5%	band	28.6	held
098 P1-Qwen325	FreeToken’s best at most 1.10× its Table 1 setting, Qwen3 25%	threshold	1.014 [1.010, 1.019]	held
098 P2-Qwen325	ours / FreeToken-best in 0.90-1.12, Qwen3 25%	band	1.044 [1.032, 1.056]	held
098 P3-Qwen325-cap1	fetch cap 1 within 8% of the calibrated split, Qwen3 25%	band	-8.0	held (point)
098 P3-Qwen325-cap2	fetch cap 2 within 8% of the calibrated split, Qwen3 25%	band	-8.7	failed
098 P4-Qwen325	8 threads slower than default or within 3%, Qwen3 25%	threshold	1.4	held (point)
098 P5-Qwen325	FreeToken’s best backend is hybrid, Qwen3 25%	equality	1	held
098 P6-Qwen325	FreeToken 27-30 GiB, ours below it, Qwen3 25%	band	28.6	held

continued

Job	Clause	Type	Measured	Status
098 P1-Qwen343.75	FreeToken's best at most 1.10× its Table 1 setting, Qwen3 43.75%	threshold	1.000 [1.000, 1.000]	held
098 P2-Qwen343.75	ours / FreeToken-best in 0.90-1.12, Qwen3 43.75%	band	0.983 [0.970, 0.996]	held
098 P3-Qwen343.75-cap1	fetch cap 1 within 8% of the calibrated split, Qwen3 43.75%	band	-5.3	held (point)
098 P3-Qwen343.75-cap2	fetch cap 2 within 8% of the calibrated split, Qwen3 43.75%	band	-6.8	held (point)
098 P4-Qwen343.75	8 threads slower than default or within 3%, Qwen3 43.75%	threshold	1.1	held (point)
098 P5-Qwen343.75	FreeToken's best backend is offload, Qwen3 43.75%	equality	1	held
098 P6-Qwen343.75	FreeToken 27-30 GiB, ours below it, Qwen3 43.75%	band	30.1	failed
098 P7-gpt-oss	CPU-only: llama.cpp at least 2.0× FreeToken's cpu backend, gpt-oss	threshold	1.128 [1.126, 1.130]	failed
098 P7-Qwen3	CPU-only: llama.cpp at least 1.2× FreeToken's cpu backend, Qwen3	threshold	0.988 [0.987, 0.989]	failed
097O4 P1	six model files hash to the replay's (O4)	equality	6	held
097O4 P2a-gpt-oss11	learned reads fewer than foa, gpt-oss 11% (O4)	sign	5.9	held (point)
097O4 P2b-gpt-oss11	learned's read reduction within 5 points of the replay's, gpt-oss 11% (O4)	band	0.1	held (point)
097O4 P3-gpt-oss11	learned 1.01-1.18× foa, gpt-oss 11% (O4)	band	1.011 [1.001, 1.020]	held (point)
097O4 P5a-gpt-oss11	learned faster than base, gpt-oss 11% (O4)	sign	1.030 [1.023, 1.039]	held
097O4 P5b-gpt-oss11	learned recovers 10-45% of fetch's gain, gpt-oss 11% (O4)	band	0.084	failed
097O4 P6-gpt-oss11	learned host time at most 300 us per token, gpt-oss 11% (O4)	threshold	315	failed
097O4 P7-foa-gpt-oss11	foa/base within 0.02 of job 096 on the same machine, gpt-oss 11% (O4)	band	-0.001	held (point)
097O4 P7-fetch-gpt-oss11	fetch/base within 0.05 of job 096 on the same machine, gpt-oss 11% (O4)	band	-0.001	held (point)
097O4 P7-both3p-gpt-oss11	both3p/base within 0.08 of job 096 on the same machine, gpt-oss 11% (O4)	band	-0.005	held (point)
097O4 P2a-gpt-oss25	learned reads fewer than foa, gpt-oss 25% (O4)	sign	11.5	held (point)
097O4 P2b-gpt-oss25	learned's read reduction within 5 points of the replay's, gpt-oss 25% (O4)	band	3.9	held (point)
097O4 P3-gpt-oss25	learned 1.01-1.18× foa, gpt-oss 25% (O4)	band	1.030 [1.018, 1.041]	held
097O4 P5a-gpt-oss25	learned faster than base, gpt-oss 25% (O4)	sign	1.053 [1.043, 1.063]	held
097O4 P5b-gpt-oss25	learned recovers 10-45% of fetch's gain, gpt-oss 25% (O4)	band	0.129	held (point)
097O4 P6-gpt-oss25	learned host time at most 300 us per token, gpt-oss 25% (O4)	threshold	286	held (point)
097O4 P7-foa-gpt-oss25	foa/base within 0.02 of job 096 on the same machine, gpt-oss 25% (O4)	band	-0.007	held (point)
097O4 P7-fetch-gpt-oss25	fetch/base within 0.05 of job 096 on the same machine, gpt-oss 25% (O4)	band	-0.013	held (point)

continued

Job	Clause	Type	Measured	Status
097O4 P7-both3p-gpt-oss25	both3p/base within 0.08 of job 096 on the same machine, gpt-oss 25% (O4)	band	-0.007	held (point)
097O4 P2a-gpt-oss40	learned reads fewer than foa, gpt-oss 40% (O4)	sign	16.1	held (point)
097O4 P2b-gpt-oss40	learned's read reduction within 5 points of the replay's, gpt-oss 40% (O4)	band	4.0	held (point)
097O4 P4-gpt-oss40	learned at least $0.99\times$ foa, gpt-oss 40% (O4)	threshold	1.045 [1.037, 1.052]	held
097O4 P6-gpt-oss40	learned host time at most 300 us per token, gpt-oss 40% (O4)	threshold	240	held (point)
097O4 P7-foa-gpt-oss40	foa/base within 0.02 of job 096 on the same machine, gpt-oss 40% (O4)	band	0.005	held (point)
097O4 P7-fetch-gpt-oss40	fetch/base within 0.05 of job 096 on the same machine, gpt-oss 40% (O4)	band	0.007	held (point)
097O4 P7-both3p-gpt-oss40	both3p/base within 0.08 of job 096 on the same machine, gpt-oss 40% (O4)	band	0.023	held (point)
097O4 P2a-Qwen312.5	learned reads fewer than foa, Qwen3 12.5% (O4)	sign	5.6	held (point)
097O4 P2b-Qwen312.5	learned's read reduction within 5 points of the replay's, Qwen3 12.5% (O4)	band	-6.4	failed
097O4 P3-Qwen312.5	learned $1.01-1.18\times$ foa, Qwen3 12.5% (O4)	band	1.018 [1.015, 1.022]	held
097O4 P5a-Qwen312.5	learned faster than base, Qwen3 12.5% (O4)	sign	1.032 [1.029, 1.036]	held
097O4 P5b-Qwen312.5	learned recovers 10-45% of fetch's gain, Qwen3 12.5% (O4)	band	0.074	failed
097O4 P6-Qwen312.5	learned host time at most 600 us per token, Qwen3 12.5% (O4)	threshold	580	held (point)
097O4 P7-foa-Qwen312.5	foa/base within 0.02 of job 096 on the same machine, Qwen3 12.5% (O4)	band	-0.001	held (point)
097O4 P7-fetch-Qwen312.5	fetch/base within 0.05 of job 096 on the same machine, Qwen3 12.5% (O4)	band	-0.002	held (point)
097O4 P7-both3p-Qwen312.5	both3p/base within 0.08 of job 096 on the same machine, Qwen3 12.5% (O4)	band	0.004	held (point)
097O4 P2a-Qwen325	learned reads fewer than foa, Qwen3 25% (O4)	sign	11.3	held (point)
097O4 P2b-Qwen325	learned's read reduction within 5 points of the replay's, Qwen3 25% (O4)	band	0.2	held (point)
097O4 P3-Qwen325	learned $1.01-1.18\times$ foa, Qwen3 25% (O4)	band	1.034 [1.026, 1.040]	held
097O4 P5a-Qwen325	learned faster than base, Qwen3 25% (O4)	sign	1.052 [1.045, 1.059]	held
097O4 P5b-Qwen325	learned recovers 10-45% of fetch's gain, Qwen3 25% (O4)	band	0.101	held (point)
097O4 P6-Qwen325	learned host time at most 600 us per token, Qwen3 25% (O4)	threshold	547	held (point)
097O4 P7-foa-Qwen325	foa/base within 0.02 of job 096 on the same machine, Qwen3 25% (O4)	band	0.002	held (point)
097O4 P7-fetch-Qwen325	fetch/base within 0.05 of job 096 on the same machine, Qwen3 25% (O4)	band	0.000	held (point)

continued

Job	Clause	Type	Measured	Status
097O4 P7-both3p-Qwen325	both3p/base within 0.08 of job 096 on the same machine, Qwen3 25% (O4)	band	-0.003	held (point)
097O4 P2a-Qwen343.75	learned reads fewer than foa, Qwen3 43.75% (O4)	sign	17.1	held (point)
097O4 P2b-Qwen343.75	learned's read reduction within 5 points of the replay's, Qwen3 43.75% (O4)	band	8.1	failed
097O4 P4-Qwen343.75	learned at least $0.99\times$ foa, Qwen3 43.75% (O4)	threshold	1.024 [1.017, 1.030]	held
097O4 P6-Qwen343.75	learned host time at most 600 us per token, Qwen3 43.75% (O4)	threshold	460	held (point)
097O4 P7-foa-Qwen343.75	foa/base within 0.02 of job 096 on the same machine, Qwen3 43.75% (O4)	band	0.003	held (point)
097O4 P7-fetch-Qwen343.75	fetch/base within 0.05 of job 096 on the same machine, Qwen3 43.75% (O4)	band	-0.003	held (point)
097O4 P7-both3p-Qwen343.75	both3p/base within 0.08 of job 096 on the same machine, Qwen3 43.75% (O4)	band	-0.006	held (point)
097O6 P1	six model files hash to the replay's (O6)	equality	6	held
097O6 P2a-gpt-oss11	learned reads fewer than foa, gpt-oss 11% (O6)	sign	6.0	held (point)
097O6 P2b-gpt-oss11	learned's read reduction within 5 points of the replay's, gpt-oss 11% (O6)	band	0.2	held (point)
097O6 P3-gpt-oss11	learned $1.01-1.18\times$ foa, gpt-oss 11% (O6)	band	0.942 [0.937, 0.947]	failed
097O6 P5a-gpt-oss11	learned faster than base, gpt-oss 11% (O6)	sign	0.936 [0.931, 0.941]	failed
097O6 P5b-gpt-oss11	learned recovers 10-45% of fetch's gain, gpt-oss 11% (O6)	band	-0.271	failed
097O6 P6-gpt-oss11	learned host time at most 300 us per token, gpt-oss 11% (O6)	threshold	292	held (point)
097O6 P2a-gpt-oss25	learned reads fewer than foa, gpt-oss 25% (O6)	sign	11.4	held (point)
097O6 P2b-gpt-oss25	learned's read reduction within 5 points of the replay's, gpt-oss 25% (O6)	band	3.8	held (point)
097O6 P3-gpt-oss25	learned $1.01-1.18\times$ foa, gpt-oss 25% (O6)	band	0.939 [0.933, 0.945]	failed
097O6 P5a-gpt-oss25	learned faster than base, gpt-oss 25% (O6)	sign	0.926 [0.920, 0.933]	failed
097O6 P5b-gpt-oss25	learned recovers 10-45% of fetch's gain, gpt-oss 25% (O6)	band	-0.283	failed
097O6 P6-gpt-oss25	learned host time at most 300 us per token, gpt-oss 25% (O6)	threshold	275	held (point)
097O6 P2a-gpt-oss40	learned reads fewer than foa, gpt-oss 40% (O6)	sign	16.0	held (point)
097O6 P2b-gpt-oss40	learned's read reduction within 5 points of the replay's, gpt-oss 40% (O6)	band	3.9	held (point)
097O6 P4-gpt-oss40	learned at least $0.99\times$ foa, gpt-oss 40% (O6)	threshold	0.969 [0.964, 0.974]	failed
097O6 P6-gpt-oss40	learned host time at most 300 us per token, gpt-oss 40% (O6)	threshold	237	held (point)
097O6 P2a-Qwen312.5	learned reads fewer than foa, Qwen3 12.5% (O6)	sign	7.7	held (point)

continued

Job	Clause	Type	Measured	Status
097O6 P2b- Qwen312.5	learned's read reduction within 5 points of the replay's, Qwen3 12.5% (O6)	band	-4.3	held (point)
097O6 P3- Qwen312.5	learned 1.01-1.18 $\times$ foa, Qwen3 12.5% (O6)	band	0.915 [0.911, 0.919]	failed
097O6 P5a- Qwen312.5	learned faster than base, Qwen3 12.5% (O6)	sign	0.910 [0.906, 0.914]	failed
097O6 P5b- Qwen312.5	learned recovers 10-45% of fetch's gain, Qwen3 12.5% (O6)	band	-0.308	failed
097O6 P6- Qwen312.5	learned host time at most 600 us per token, Qwen3 12.5% (O6)	threshold	525	held (point)
097O6 P2a- Qwen325	learned reads fewer than foa, Qwen3 25% (O6)	sign	13.2	held (point)
097O6 P2b- Qwen325	learned's read reduction within 5 points of the replay's, Qwen3 25% (O6)	band	2.1	held (point)
097O6 P3- Qwen325	learned 1.01-1.18 $\times$ foa, Qwen3 25% (O6)	band	0.905 [0.899, 0.911]	failed
097O6 P5a- Qwen325	learned faster than base, Qwen3 25% (O6)	sign	0.896 [0.890, 0.901]	failed
097O6 P5b- Qwen325	learned recovers 10-45% of fetch's gain, Qwen3 25% (O6)	band	-0.342	failed
097O6 P6- Qwen325	learned host time at most 600 us per token, Qwen3 25% (O6)	threshold	585	held (point)
097O6 P2a- Qwen343.75	learned reads fewer than foa, Qwen3 43.75% (O6)	sign	17.9	held (point)
097O6 P2b- Qwen343.75	learned's read reduction within 5 points of the replay's, Qwen3 43.75% (O6)	band	8.9	failed
097O6 P4- Qwen343.75	learned at least 0.99 $\times$ foa, Qwen3 43.75% (O6)	threshold	0.933 [0.927, 0.939]	failed
097O6 P6- Qwen343.75	learned host time at most 600 us per token, Qwen3 43.75% (O6)	threshold	483	held (point)

Table 2: Job 099 (the host panel), scored by machine by `scripts/panel.099.py` under the same rule. Totals: 273 clauses, 190 held, 42 held (point), 39 failed, 2 untested. Hosts Pa-Pj; cells gpt-oss 11% (g11) and 25% (g25), Qwen3 12.5% (P10). Measured: the point estimate and, where one exists, the 95% interval (bootstrap over problems within the host; pooled clauses have none).

Clause	Statement	Measured	Predicted	Status
Pa-P1-g11	engine reads within 4% of the replay (gpt-oss 11%, Pa)	0.016	≤ 0.04	held (point)
Pa-P3-w16-g11	w16 time share (gpt-oss 11%, Pa)	0.886 [0.880, 0.892]	≥ 0.80	held
Pa-P3-w4-g11	w4 time share (gpt-oss 11%, Pa)	0.426 [0.415, 0.438]	0.35-0.70	held
Pa-P3-w1-g11	w1 time share (gpt-oss 11%, Pa)	0.167 [0.160, 0.175]	≤ 0.30	held
Pa-P4-w8r5-g11	w8r5 time share (gpt-oss 11%, Pa)	0.240 [0.226, 0.253]	0.15-0.45	held
Pa-P4-allr5-g11	allr5 time share (gpt-oss 11%, Pa)	0.455 [0.442, 0.467]	0.30-0.60	held
Pa-P5a-g11	MIN's set worth more with one read than two (gpt-oss 11%, Pa)	3.873 [3.717, 4.038]	> 0 ms	held
Pa-P5b-g11	single read within 5% of deployed (gpt-oss 11%, Pa)	1.004 [1.003, 1.006]	0.95-1.05	held
Pa-P6-fetch-g11	fetch/base (gpt-oss 11%, Pa)	1.334 [1.325, 1.343]	1.1-1.55	held
Pa-P6-both3p-g11	both3p/base (gpt-oss 11%, Pa)	1.515 [1.501, 1.528]	1.25-1.9	held
Pa-P6-order-g11	both3p $>$ fetch $>$ foa (gpt-oss 11%, Pa)	1.000	ordering	held (point)
Pa-P9-g11	host plan at most 150 μ s per step (gpt-oss 11%, Pa)	261.2	$\leq 150 \mu$ s	failed
Pa-P7-g11	aa / foa in [0.92, 1.04] on a ≥ 40 GB/s link (gpt-oss 11%, Pa)	0.901 [0.899, 0.904]	0.92-1.04	failed
Pa-P1-g25	engine reads within 4% of the replay (gpt-oss 25%, Pa)	0.017	≤ 0.04	held (point)
Pa-P3-w16-g25	w16 time share (gpt-oss 25%, Pa)	0.573 [0.564, 0.582]	0.45-0.85	held
Pa-P3-w4-g25	w4 time share (gpt-oss 25%, Pa)	0.235 [0.230, 0.240]	0.12-0.40	held
Pa-P3-w1-g25	w1 time share (gpt-oss 25%, Pa)	0.084 [0.081, 0.088]	≤ 0.30	held
Pa-P4-w8r5-g25	w8r5 time share (gpt-oss 25%, Pa)	0.140 [0.136, 0.145]	≤ 0.30	held
Pa-P4-allr5-g25	allr5 time share (gpt-oss 25%, Pa)	0.402 [0.392, 0.413]	0.30-0.60	held
Pa-P5a-g25	MIN's set worth more with one read than two (gpt-oss 25%, Pa)	1.203 [1.127, 1.277]	> 0 ms	held
Pa-P5b-g25	single read within 5% of deployed (gpt-oss 25%, Pa)	0.997 [0.995, 0.999]	0.95-1.05	held
Pa-P6-fetch-g25	fetch/base (gpt-oss 25%, Pa)	1.332 [1.318, 1.344]	1.1-1.55	held
Pa-P6-both3p-g25	both3p/base (gpt-oss 25%, Pa)	1.687 [1.661, 1.710]	1.25-1.9	held
Pa-P6-order-g25	both3p $>$ fetch $>$ foa (gpt-oss 25%, Pa)	1.000	ordering	held (point)
Pa-P9-g25	host plan at most 150 μ s per step (gpt-oss 25%, Pa)	258.4	$\leq 150 \mu$ s	failed
Pa-P7-g25	aa / foa in [0.92, 1.04] on a ≥ 40 GB/s link (gpt-oss 25%, Pa)	0.928 [0.924, 0.932]	0.92-1.04	held
Pa-P10-w2	Qwen3 w2 time share (Pa)	0.414 [0.403, 0.425]	0.30-0.70	held
Pa-P10-w8r5	Qwen3 w8r5 time share (Pa)	0.382 [0.367, 0.395]	0.25-0.60	held
Pa-P10-fetch	Qwen3 fetch/base (Pa)	1.419 [1.410, 1.428]	1.15-1.55	held
Pb-P1-g11	engine reads within 4% of the replay (gpt-oss 11%, Pb)	0.016	≤ 0.04	held (point)
Pb-P3-w16-g11	w16 time share (gpt-oss 11%, Pb)	0.928 [0.923, 0.932]	≥ 0.80	held
Pb-P3-w4-g11	w4 time share (gpt-oss 11%, Pb)	0.468 [0.456, 0.480]	0.35-0.70	held
Pb-P3-w1-g11	w1 time share (gpt-oss 11%, Pb)	0.171 [0.159, 0.184]	≤ 0.30	held
Pb-P4-w8r5-g11	w8r5 time share (gpt-oss 11%, Pb)	0.265 [0.254, 0.276]	0.15-0.45	held
Pb-P4-allr5-g11	allr5 time share (gpt-oss 11%, Pb)	0.438 [0.422, 0.453]	0.30-0.60	held
Pb-P5a-g11	MIN's set worth more with one read than two (gpt-oss 11%, Pb)	4.823 [4.654, 4.997]	> 0 ms	held
Pb-P5b-g11	single read within 5% of deployed (gpt-oss 11%, Pb)	1.005 [1.003, 1.007]	0.95-1.05	held
Pb-P6-fetch-g11	fetch/base (gpt-oss 11%, Pb)	1.343 [1.334, 1.352]	1.1-1.55	held
Pb-P6-both3p-g11	both3p/base (gpt-oss 11%, Pb)	1.456 [1.444, 1.469]	1.25-1.9	held
Pb-P6-order-g11	both3p $>$ fetch $>$ foa (gpt-oss 11%, Pb)	1.000	ordering	held (point)
Pb-P9-g11	host plan at most 150 μ s per step (gpt-oss 11%, Pb)	202.5	$\leq 150 \mu$ s	failed
Pb-P7-g11	aa / foa in [0.92, 1.04] on a ≥ 40 GB/s link (gpt-oss 11%, Pb)	0.955 [0.952, 0.958]	0.92-1.04	held
Pb-P1-g25	engine reads within 4% of the replay (gpt-oss 25%, Pb)	0.016	≤ 0.04	held (point)
Pb-P3-w16-g25	w16 time share (gpt-oss 25%, Pb)	0.620 [0.612, 0.629]	0.45-0.85	held
Pb-P3-w4-g25	w4 time share (gpt-oss 25%, Pb)	0.240 [0.234, 0.245]	0.12-0.40	held
Pb-P3-w1-g25	w1 time share (gpt-oss 25%, Pb)	0.083 [0.079, 0.087]	≤ 0.30	held
Pb-P4-w8r5-g25	w8r5 time share (gpt-oss 25%, Pb)	0.141 [0.136, 0.147]	≤ 0.30	held
Pb-P4-allr5-g25	allr5 time share (gpt-oss 25%, Pb)	0.420 [0.409, 0.431]	0.30-0.60	held
Pb-P5a-g25	MIN's set worth more with one read than two (gpt-oss 25%, Pb)	1.605 [1.502, 1.712]	> 0 ms	held
Pb-P5b-g25	single read within 5% of deployed (gpt-oss 25%, Pb)	1.016 [1.012, 1.018]	0.95-1.05	held
Pb-P6-fetch-g25	fetch/base (gpt-oss 25%, Pb)	1.403 [1.387, 1.419]	1.1-1.55	held
Pb-P6-both3p-g25	both3p/base (gpt-oss 25%, Pb)	1.713 [1.690, 1.735]	1.25-1.9	held

continued

Clause	Statement	Measured	Predicted	Status
Pb-P6-order-g25	both3p > fetch > foa (gpt-oss 25%, Pb)	1.000	ordering	held (point)
Pb-P9-g25	host plan at most 150 μ s per step (gpt-oss 25%, Pb)	200.9	$\leq 150 \mu$ s	failed
Pb-P7-g25	aa / foa in [0.92, 1.04] on a ≥ 40 GB/s link (gpt-oss 25%, Pb)	0.974 [0.969, 0.980]	0.92-1.04	held
Pb-P10-w2	Qwen3 w2 time share (Pb)	0.449 [0.438, 0.459]	0.30-0.70	held
Pb-P10-w8r5	Qwen3 w8r5 time share (Pb)	0.391 [0.378, 0.404]	0.25-0.60	held
Pb-P10-fetch	Qwen3 fetch/base (Pb)	1.439 [1.430, 1.448]	1.15-1.55	held
Pc-P1-g11	engine reads within 4% of the replay (gpt-oss 11%, Pc)	0.016	≤ 0.04	held (point)
Pc-P3-w16-g11	w16 time share (gpt-oss 11%, Pc)	0.844 [0.836, 0.851]	≥ 0.80	held
Pc-P3-w4-g11	w4 time share (gpt-oss 11%, Pc)	0.362 [0.351, 0.375]	0.35-0.70	held
Pc-P3-w1-g11	w1 time share (gpt-oss 11%, Pc)	0.156 [0.149, 0.163]	≤ 0.30	held
Pc-P4-w8r5-g11	w8r5 time share (gpt-oss 11%, Pc)	0.222 [0.212, 0.231]	0.15-0.45	held
Pc-P4-allr5-g11	allr5 time share (gpt-oss 11%, Pc)	0.549 [0.540, 0.558]	0.30-0.60	held
Pc-P5a-g11	MIN's set worth more with one read than two (gpt-oss 11%, Pc)	3.322 [3.167, 3.483]	> 0 ms	held
Pc-P5b-g11	single read within 5% of deployed (gpt-oss 11%, Pc)	0.996 [0.992, 1.000]	0.95-1.05	held
Pc-P6-fetch-g11	fetch/base (gpt-oss 11%, Pc)	1.168 [1.160, 1.176]	1.1-1.55	held
Pc-P6-both3p-g11	both3p/base (gpt-oss 11%, Pc)	1.328 [1.316, 1.341]	1.25-1.9	held
Pc-P6-order-g11	both3p > fetch > foa (gpt-oss 11%, Pc)	1.000	ordering	held (point)
Pc-P9-g11	host plan at most 150 μ s per step (gpt-oss 11%, Pc)	202.4	$\leq 150 \mu$ s	failed
Pc-P7-g11	aa / foa below 0.95 on a < 32 GB/s link (gpt-oss 11%, Pc)	0.657 [0.654, 0.660]	< 0.95	held
Pc-P1-g25	engine reads within 4% of the replay (gpt-oss 25%, Pc)	0.016	≤ 0.04	held (point)
Pc-P3-w16-g25	w16 time share (gpt-oss 25%, Pc)	0.540 [0.533, 0.549]	0.45-0.85	held
Pc-P3-w4-g25	w4 time share (gpt-oss 25%, Pc)	0.228 [0.222, 0.235]	0.12-0.40	held
Pc-P3-w1-g25	w1 time share (gpt-oss 25%, Pc)	0.099 [0.094, 0.105]	≤ 0.30	held
Pc-P4-w8r5-g25	w8r5 time share (gpt-oss 25%, Pc)	0.149 [0.144, 0.155]	≤ 0.30	held
Pc-P4-allr5-g25	allr5 time share (gpt-oss 25%, Pc)	0.483 [0.474, 0.492]	0.30-0.60	held
Pc-P5a-g25	MIN's set worth more with one read than two (gpt-oss 25%, Pc)	0.974 [0.892, 1.065]	> 0 ms	held
Pc-P5b-g25	single read within 5% of deployed (gpt-oss 25%, Pc)	0.974 [0.969, 0.978]	0.95-1.05	held
Pc-P6-fetch-g25	fetch/base (gpt-oss 25%, Pc)	1.237 [1.224, 1.249]	1.1-1.55	held
Pc-P6-both3p-g25	both3p/base (gpt-oss 25%, Pc)	1.536 [1.521, 1.551]	1.25-1.9	held
Pc-P6-order-g25	both3p > fetch > foa (gpt-oss 25%, Pc)	1.000	ordering	held (point)
Pc-P9-g25	host plan at most 150 μ s per step (gpt-oss 25%, Pc)	218.0	$\leq 150 \mu$ s	failed
Pc-P7-g25	aa / foa below 0.95 on a < 32 GB/s link (gpt-oss 25%, Pc)	0.751 [0.745, 0.758]	< 0.95	held
Pc-P10-w2	Qwen3 w2 time share (Pc)	0.352 [0.343, 0.362]	0.30-0.70	held
Pc-P10-w8r5	Qwen3 w8r5 time share (Pc)	0.375 [0.367, 0.384]	0.25-0.60	held
Pc-P10-fetch	Qwen3 fetch/base (Pc)	1.211 [1.203, 1.218]	1.15-1.55	held
Pd-P1-g11	engine reads within 4% of the replay (gpt-oss 11%, Pd)	–	≤ 0.04	untested
Pd-P3-w16-g11	w16 time share (gpt-oss 11%, Pd)	0.803 [0.794, 0.812]	≥ 0.80	held (point)
Pd-P3-w4-g11	w4 time share (gpt-oss 11%, Pd)	0.342 [0.331, 0.354]	0.35-0.70	failed
Pd-P3-w1-g11	w1 time share (gpt-oss 11%, Pd)	0.147 [0.140, 0.154]	≤ 0.30	held
Pd-P4-w8r5-g11	w8r5 time share (gpt-oss 11%, Pd)	0.209 [0.199, 0.219]	0.15-0.45	held
Pd-P4-allr5-g11	allr5 time share (gpt-oss 11%, Pd)	0.521 [0.511, 0.532]	0.30-0.60	held
Pd-P5a-g11	MIN's set worth more with one read than two (gpt-oss 11%, Pd)	1.169 [1.031, 1.315]	> 0 ms	held
Pd-P5b-g11	single read within 5% of deployed (gpt-oss 11%, Pd)	0.981 [0.975, 0.987]	0.95-1.05	held
Pd-P6-fetch-g11	fetch/base (gpt-oss 11%, Pd)	1.051 [1.043, 1.058]	1.1-1.55	failed
Pd-P6-both3p-g11	both3p/base (gpt-oss 11%, Pd)	1.112 [1.105, 1.120]	1.25-1.9	failed
Pd-P6-order-g11	both3p > fetch > foa (gpt-oss 11%, Pd)	1.000	ordering	held (point)
Pd-P9-g11	host plan at most 150 μ s per step (gpt-oss 11%, Pd)	233.7	$\leq 150 \mu$ s	failed
Pd-P7-g11	aa / foa below 0.95 on a < 32 GB/s link (gpt-oss 11%, Pd)	0.566 [0.562, 0.570]	< 0.95	held
Pd-P1-g25	engine reads within 4% of the replay (gpt-oss 25%, Pd)	–	≤ 0.04	untested
Pd-P3-w16-g25	w16 time share (gpt-oss 25%, Pd)	0.522 [0.512, 0.532]	0.45-0.85	held
Pd-P3-w4-g25	w4 time share (gpt-oss 25%, Pd)	0.221 [0.213, 0.231]	0.12-0.40	held
Pd-P3-w1-g25	w1 time share (gpt-oss 25%, Pd)	0.099 [0.092, 0.107]	≤ 0.30	held
Pd-P4-w8r5-g25	w8r5 time share (gpt-oss 25%, Pd)	0.147 [0.140, 0.154]	≤ 0.30	held
Pd-P4-allr5-g25	allr5 time share (gpt-oss 25%, Pd)	0.435 [0.425, 0.445]	0.30-0.60	held
Pd-P5a-g25	MIN's set worth more with one read than two (gpt-oss 25%, Pd)	0.261 [0.101, 0.421]	> 0 ms	held
Pd-P5b-g25	single read within 5% of deployed (gpt-oss 25%, Pd)	0.922 [0.908, 0.936]	0.95-1.05	failed
Pd-P6-fetch-g25	fetch/base (gpt-oss 25%, Pd)	1.124 [1.108, 1.139]	1.1-1.55	held

continued

Clause	Statement	Measured	Predicted	Status
Pd-P6-both3p-g25	both3p/base (gpt-oss 25%, Pd)	1.315 [1.299, 1.331]	1.25-1.9	held
Pd-P6-order-g25	both3p > fetch > foa (gpt-oss 25%, Pd)	1.000	ordering	held (point)
Pd-P9-g25	host plan at most 150 μ s per step (gpt-oss 25%, Pd)	238.2	$\leq 150 \mu$ s	failed
Pd-P7-g25	aa / foa below 0.95 on a < 32 GB/s link (gpt-oss 25%, Pd)	0.692 [0.683, 0.701]	< 0.95	held
Pe-P1-g11	engine reads within 4% of the replay (gpt-oss 11%, Pe)	0.016	≤ 0.04	held (point)
Pe-P3-w16-g11	w16 time share (gpt-oss 11%, Pe)	0.935 [0.930, 0.939]	≥ 0.80	held
Pe-P3-w4-g11	w4 time share (gpt-oss 11%, Pe)	0.469 [0.457, 0.481]	0.35-0.70	held
Pe-P3-w1-g11	w1 time share (gpt-oss 11%, Pe)	0.164 [0.156, 0.173]	≤ 0.30	held
Pe-P4-w8r5-g11	w8r5 time share (gpt-oss 11%, Pe)	0.257 [0.246, 0.267]	0.15-0.45	held
Pe-P4-allr5-g11	allr5 time share (gpt-oss 11%, Pe)	0.415 [0.400, 0.430]	0.30-0.60	held
Pe-P5a-g11	MIN's set worth more with one read than two (gpt-oss 11%, Pe)	4.784 [4.621, 4.957]	> 0 ms	held
Pe-P5b-g11	single read within 5% of deployed (gpt-oss 11%, Pe)	1.008 [1.007, 1.009]	0.95-1.05	held
Pe-P6-fetch-g11	fetch/base (gpt-oss 11%, Pe)	1.348 [1.338, 1.357]	1.1-1.55	held
Pe-P6-both3p-g11	both3p/base (gpt-oss 11%, Pe)	1.461 [1.447, 1.474]	1.25-1.9	held
Pe-P6-order-g11	both3p > fetch > foa (gpt-oss 11%, Pe)	1.000	ordering	held (point)
Pe-P9-g11	host plan at most 150 μ s per step (gpt-oss 11%, Pe)	262.4	$\leq 150 \mu$ s	failed
Pe-P7-g11	aa / foa in [0.92, 1.04] on a ≥ 40 GB/s link (gpt-oss 11%, Pe)	0.966 [0.963, 0.969]	0.92-1.04	held
Pe-P1-g25	engine reads within 4% of the replay (gpt-oss 25%, Pe)	0.016	≤ 0.04	held (point)
Pe-P3-w16-g25	w16 time share (gpt-oss 25%, Pe)	0.631 [0.623, 0.640]	0.45-0.85	held
Pe-P3-w4-g25	w4 time share (gpt-oss 25%, Pe)	0.238 [0.232, 0.243]	0.12-0.40	held
Pe-P3-w1-g25	w1 time share (gpt-oss 25%, Pe)	0.071 [0.065, 0.076]	≤ 0.30	held
Pe-P4-w8r5-g25	w8r5 time share (gpt-oss 25%, Pe)	0.135 [0.131, 0.139]	≤ 0.30	held
Pe-P4-allr5-g25	allr5 time share (gpt-oss 25%, Pe)	0.405 [0.393, 0.416]	0.30-0.60	held
Pe-P5a-g25	MIN's set worth more with one read than two (gpt-oss 25%, Pe)	1.637 [1.529, 1.749]	> 0 ms	held
Pe-P5b-g25	single read within 5% of deployed (gpt-oss 25%, Pe)	1.014 [1.011, 1.017]	0.95-1.05	held
Pe-P6-fetch-g25	fetch/base (gpt-oss 25%, Pe)	1.403 [1.387, 1.419]	1.1-1.55	held
Pe-P6-both3p-g25	both3p/base (gpt-oss 25%, Pe)	1.710 [1.687, 1.734]	1.25-1.9	held
Pe-P6-order-g25	both3p > fetch > foa (gpt-oss 25%, Pe)	1.000	ordering	held (point)
Pe-P9-g25	host plan at most 150 μ s per step (gpt-oss 25%, Pe)	258.2	$\leq 150 \mu$ s	failed
Pe-P7-g25	aa / foa in [0.92, 1.04] on a ≥ 40 GB/s link (gpt-oss 25%, Pe)	0.988 [0.983, 0.993]	0.92-1.04	held
Pf-P1-g11	engine reads within 4% of the replay (gpt-oss 11%, Pf)	0.016	≤ 0.04	held (point)
Pf-P3-w16-g11	w16 time share (gpt-oss 11%, Pf)	0.781 [0.771, 0.791]	≥ 0.80	failed
Pf-P3-w4-g11	w4 time share (gpt-oss 11%, Pf)	0.320 [0.309, 0.331]	0.35-0.70	failed
Pf-P3-w1-g11	w1 time share (gpt-oss 11%, Pf)	0.141 [0.134, 0.148]	≤ 0.30	held
Pf-P4-w8r5-g11	w8r5 time share (gpt-oss 11%, Pf)	0.198 [0.188, 0.208]	0.15-0.45	held
Pf-P4-allr5-g11	allr5 time share (gpt-oss 11%, Pf)	0.540 [0.530, 0.549]	0.30-0.60	held
Pf-P5a-g11	MIN's set worth more with one read than two (gpt-oss 11%, Pf)	-0.484 [-1.055, -0.139]	> 0 ms	failed
Pf-P5b-g11	single read within 5% of deployed (gpt-oss 11%, Pf)	0.947 [0.928, 0.983]	0.95-1.05	failed
Pf-P6-fetch-g11	fetch/base (gpt-oss 11%, Pf)	0.875 [0.855, 0.911]	1.1-1.55	failed
Pf-P6-both3p-g11	both3p/base (gpt-oss 11%, Pf)	1.008 [0.975, 1.056]	1.25-1.9	failed
Pf-P6-order-g11	both3p > fetch > foa (gpt-oss 11%, Pf)	0.000	ordering	failed
Pf-P9-g11	host plan at most 150 μ s per step (gpt-oss 11%, Pf)	201.3	$\leq 150 \mu$ s	failed
Pf-P7-g11	aa / foa below 0.95 on a < 32 GB/s link (gpt-oss 11%, Pf)	0.448 [0.444, 0.452]	< 0.95	held
Pf-P1-g25	engine reads within 4% of the replay (gpt-oss 25%, Pf)	0.017	≤ 0.04	held (point)
Pf-P3-w16-g25	w16 time share (gpt-oss 25%, Pf)	0.493 [0.485, 0.502]	0.45-0.85	held
Pf-P3-w4-g25	w4 time share (gpt-oss 25%, Pf)	0.213 [0.207, 0.221]	0.12-0.40	held
Pf-P3-w1-g25	w1 time share (gpt-oss 25%, Pf)	0.095 [0.089, 0.102]	≤ 0.30	held
Pf-P4-w8r5-g25	w8r5 time share (gpt-oss 25%, Pf)	0.140 [0.134, 0.148]	≤ 0.30	held
Pf-P4-allr5-g25	allr5 time share (gpt-oss 25%, Pf)	0.434 [0.425, 0.444]	0.30-0.60	held
Pf-P5a-g25	MIN's set worth more with one read than two (gpt-oss 25%, Pf)	0.192 [0.143, 0.243]	> 0 ms	held
Pf-P5b-g25	single read within 5% of deployed (gpt-oss 25%, Pf)	0.853 [0.848, 0.859]	0.95-1.05	failed
Pf-P6-fetch-g25	fetch/base (gpt-oss 25%, Pf)	0.969 [0.964, 0.975]	1.1-1.55	failed
Pf-P6-both3p-g25	both3p/base (gpt-oss 25%, Pf)	1.202 [1.194, 1.211]	1.25-1.9	failed
Pf-P6-order-g25	both3p > fetch > foa (gpt-oss 25%, Pf)	1.000	ordering	held (point)
Pf-P9-g25	host plan at most 150 μ s per step (gpt-oss 25%, Pf)	198.6	$\leq 150 \mu$ s	failed

continued

Clause	Statement	Measured	Predicted	Status
Pf-P7-g25	aa / foa below 0.95 on a < 32 GB/s link (gpt-oss 25%, Pf)	0.612 [0.603, 0.622]	< 0.95	held
Pg-P1-g11	engine reads within 4% of the replay (gpt-oss 11%, Pg)	0.016	≤ 0.04	held (point)
Pg-P3-w16-g11	w16 time share (gpt-oss 11%, Pg)	0.835 [0.828, 0.842]	≥ 0.80	held
Pg-P3-w4-g11	w4 time share (gpt-oss 11%, Pg)	0.378 [0.367, 0.390]	0.35-0.70	held
Pg-P3-w1-g11	w1 time share (gpt-oss 11%, Pg)	0.155 [0.148, 0.162]	≤ 0.30	held
Pg-P4-w8r5-g11	w8r5 time share (gpt-oss 11%, Pg)	0.225 [0.214, 0.236]	0.15-0.45	held
Pg-P4-allr5-g11	allr5 time share (gpt-oss 11%, Pg)	0.510 [0.500, 0.521]	0.30-0.60	held
Pg-P5a-g11	MIN's set worth more with one read than two (gpt-oss 11%, Pg)	4.321 [4.105, 4.552]	> 0 ms	held
Pg-P5b-g11	single read within 5% of deployed (gpt-oss 11%, Pg)	1.012 [1.006, 1.018]	0.95-1.05	held
Pg-P6-fetch-g11	fetch/base (gpt-oss 11%, Pg)	1.245 [1.235, 1.256]	1.1-1.55	held
Pg-P6-both3p-g11	both3p/base (gpt-oss 11%, Pg)	1.381 [1.369, 1.393]	1.25-1.9	held
Pg-P6-order-g11	both3p > fetch > foa (gpt-oss 11%, Pg)	1.000	ordering	held (point)
Pg-P9-g11	host plan at most 150 μ s per step (gpt-oss 11%, Pg)	277.3	$\leq 150 \mu$ s	failed
Pg-P7-g11	aa / foa below 0.95 on a < 32 GB/s link (gpt-oss 11%, Pg)	0.722 [0.717, 0.726]	< 0.95	held
Pg-P1-g25	engine reads within 4% of the replay (gpt-oss 25%, Pg)	0.017	≤ 0.04	held (point)
Pg-P3-w16-g25	w16 time share (gpt-oss 25%, Pg)	0.544 [0.536, 0.553]	0.45-0.85	held
Pg-P3-w4-g25	w4 time share (gpt-oss 25%, Pg)	0.225 [0.219, 0.231]	0.12-0.40	held
Pg-P3-w1-g25	w1 time share (gpt-oss 25%, Pg)	0.090 [0.085, 0.095]	≤ 0.30	held
Pg-P4-w8r5-g25	w8r5 time share (gpt-oss 25%, Pg)	0.141 [0.136, 0.147]	≤ 0.30	held
Pg-P4-allr5-g25	allr5 time share (gpt-oss 25%, Pg)	0.438 [0.428, 0.448]	0.30-0.60	held
Pg-P5a-g25	MIN's set worth more with one read than two (gpt-oss 25%, Pg)	1.173 [1.090, 1.251]	> 0 ms	held
Pg-P5b-g25	single read within 5% of deployed (gpt-oss 25%, Pg)	0.997 [0.992, 1.002]	0.95-1.05	held
Pg-P6-fetch-g25	fetch/base (gpt-oss 25%, Pg)	1.320 [1.307, 1.332]	1.1-1.55	held
Pg-P6-both3p-g25	both3p/base (gpt-oss 25%, Pg)	1.630 [1.614, 1.648]	1.25-1.9	held
Pg-P6-order-g25	both3p > fetch > foa (gpt-oss 25%, Pg)	1.000	ordering	held (point)
Pg-P9-g25	host plan at most 150 μ s per step (gpt-oss 25%, Pg)	242.3	$\leq 150 \mu$ s	failed
Pg-P7-g25	aa / foa below 0.95 on a < 32 GB/s link (gpt-oss 25%, Pg)	0.794 [0.788, 0.801]	< 0.95	held
Ph-P1-g11	engine reads within 4% of the replay (gpt-oss 11%, Ph)	0.016	≤ 0.04	held (point)
Ph-P3-w16-g11	w16 time share (gpt-oss 11%, Ph)	0.887 [0.881, 0.892]	≥ 0.80	held
Ph-P3-w4-g11	w4 time share (gpt-oss 11%, Ph)	0.426 [0.415, 0.437]	0.35-0.70	held
Ph-P3-w1-g11	w1 time share (gpt-oss 11%, Ph)	0.172 [0.165, 0.179]	≤ 0.30	held
Ph-P4-w8r5-g11	w8r5 time share (gpt-oss 11%, Ph)	0.251 [0.241, 0.260]	0.15-0.45	held
Ph-P4-allr5-g11	allr5 time share (gpt-oss 11%, Ph)	0.499 [0.488, 0.510]	0.30-0.60	held
Ph-P5a-g11	MIN's set worth more with one read than two (gpt-oss 11%, Ph)	3.023 [2.907, 3.150]	> 0 ms	held
Ph-P5b-g11	single read within 5% of deployed (gpt-oss 11%, Ph)	0.991 [0.990, 0.993]	0.95-1.05	held
Ph-P6-fetch-g11	fetch/base (gpt-oss 11%, Ph)	1.246 [1.238, 1.254]	1.1-1.55	held
Ph-P6-both3p-g11	both3p/base (gpt-oss 11%, Ph)	1.424 [1.410, 1.437]	1.25-1.9	held
Ph-P6-order-g11	both3p > fetch > foa (gpt-oss 11%, Ph)	1.000	ordering	held (point)
Ph-P9-g11	host plan at most 150 μ s per step (gpt-oss 11%, Ph)	216.2	$\leq 150 \mu$ s	failed
Ph-P7-g11	aa / foa in [0.92, 1.04] on a ≥ 40 GB/s link (gpt-oss 11%, Ph)	0.845 [0.842, 0.848]	0.92-1.04	failed
Ph-P1-g25	engine reads within 4% of the replay (gpt-oss 25%, Ph)	0.016	≤ 0.04	held (point)
Ph-P3-w16-g25	w16 time share (gpt-oss 25%, Ph)	0.590 [0.581, 0.600]	0.45-0.85	held
Ph-P3-w4-g25	w4 time share (gpt-oss 25%, Ph)	0.242 [0.236, 0.247]	0.12-0.40	held
Ph-P3-w1-g25	w1 time share (gpt-oss 25%, Ph)	0.097 [0.092, 0.102]	≤ 0.30	held
Ph-P4-w8r5-g25	w8r5 time share (gpt-oss 25%, Ph)	0.144 [0.138, 0.151]	≤ 0.30	held
Ph-P4-allr5-g25	allr5 time share (gpt-oss 25%, Ph)	0.434 [0.425, 0.444]	0.30-0.60	held
Ph-P5a-g25	MIN's set worth more with one read than two (gpt-oss 25%, Ph)	1.095 [0.963, 1.224]	> 0 ms	held
Ph-P5b-g25	single read within 5% of deployed (gpt-oss 25%, Ph)	0.990 [0.986, 0.995]	0.95-1.05	held
Ph-P6-fetch-g25	fetch/base (gpt-oss 25%, Ph)	1.286 [1.274, 1.298]	1.1-1.55	held
Ph-P6-both3p-g25	both3p/base (gpt-oss 25%, Ph)	1.632 [1.609, 1.656]	1.25-1.9	held
Ph-P6-order-g25	both3p > fetch > foa (gpt-oss 25%, Ph)	1.000	ordering	held (point)
Ph-P9-g25	host plan at most 150 μ s per step (gpt-oss 25%, Ph)	202.4	$\leq 150 \mu$ s	failed
Ph-P7-g25	aa / foa in [0.92, 1.04] on a ≥ 40 GB/s link (gpt-oss 25%, Ph)	0.897 [0.893, 0.902]	0.92-1.04	failed
Pi-P1-g11	engine reads within 4% of the replay (gpt-oss 11%, Pi)	0.016	≤ 0.04	held (point)
Pi-P3-w16-g11	w16 time share (gpt-oss 11%, Pi)	0.844 [0.837, 0.851]	≥ 0.80	held
Pi-P3-w4-g11	w4 time share (gpt-oss 11%, Pi)	0.387 [0.375, 0.398]	0.35-0.70	held

continued

Clause	Statement	Measured	Predicted	Status
Pi-P3-w1-g11	w1 time share (gpt-oss 11%, Pi)	0.155 [0.147, 0.163]	≤ 0.30	held
Pi-P4-w8r5-g11	w8r5 time share (gpt-oss 11%, Pi)	0.226 [0.214, 0.237]	0.15-0.45	held
Pi-P4-allr5-g11	allr5 time share (gpt-oss 11%, Pi)	0.501 [0.489, 0.511]	0.30-0.60	held
Pi-P5a-g11	MIN's set worth more with one read than two (gpt-oss 11%, Pi)	8.898 [8.535, 9.306]	> 0 ms	held
Pi-P5b-g11	single read within 5% of deployed (gpt-oss 11%, Pi)	1.020 [1.019, 1.022]	0.95-1.05	held
Pi-P6-fetch-g11	fetch/base (gpt-oss 11%, Pi)	1.344 [1.335, 1.352]	1.1-1.55	held
Pi-P6-both3p-g11	both3p/base (gpt-oss 11%, Pi)	1.310 [1.300, 1.320]	1.25-1.9	held
Pi-P6-order-g11	both3p $>$ fetch $>$ foa (gpt-oss 11%, Pi)	0.000	ordering	failed
Pi-P9-g11	host plan at most 150 μ s per step (gpt-oss 11%, Pi)	207.8	≤ 150 μ s	failed
Pi-P7-g11	aa / foa below 0.95 on a $<$ 32 GB/s link (gpt-oss 11%, Pi)	0.766 [0.763, 0.769]	< 0.95	held
Pi-P1-g25	engine reads within 4% of the replay (gpt-oss 25%, Pi)	0.017	≤ 0.04	held (point)
Pi-P3-w16-g25	w16 time share (gpt-oss 25%, Pi)	0.558 [0.550, 0.567]	0.45-0.85	held
Pi-P3-w4-g25	w4 time share (gpt-oss 25%, Pi)	0.226 [0.221, 0.232]	0.12-0.40	held
Pi-P3-w1-g25	w1 time share (gpt-oss 25%, Pi)	0.086 [0.082, 0.091]	≤ 0.30	held
Pi-P4-w8r5-g25	w8r5 time share (gpt-oss 25%, Pi)	0.139 [0.135, 0.143]	≤ 0.30	held
Pi-P4-allr5-g25	allr5 time share (gpt-oss 25%, Pi)	0.446 [0.437, 0.454]	0.30-0.60	held
Pi-P5a-g25	MIN's set worth more with one read than two (gpt-oss 25%, Pi)	2.382 [2.231, 2.539]	> 0 ms	held
Pi-P5b-g25	single read within 5% of deployed (gpt-oss 25%, Pi)	1.018 [1.014, 1.022]	0.95-1.05	held
Pi-P6-fetch-g25	fetch/base (gpt-oss 25%, Pi)	1.444 [1.428, 1.459]	1.1-1.55	held
Pi-P6-both3p-g25	both3p/base (gpt-oss 25%, Pi)	1.578 [1.559, 1.597]	1.25-1.9	held
Pi-P6-order-g25	both3p $>$ fetch $>$ foa (gpt-oss 25%, Pi)	1.000	ordering	held (point)
Pi-P9-g25	host plan at most 150 μ s per step (gpt-oss 25%, Pi)	201.3	≤ 150 μ s	failed
Pi-P7-g25	aa / foa below 0.95 on a $<$ 32 GB/s link (gpt-oss 25%, Pi)	0.820 [0.813, 0.827]	< 0.95	held
Pj-P1-g11	engine reads within 4% of the replay (gpt-oss 11%, Pj)	0.016	≤ 0.04	held (point)
Pj-P3-w16-g11	w16 time share (gpt-oss 11%, Pj)	0.905 [0.900, 0.910]	≥ 0.80	held
Pj-P3-w4-g11	w4 time share (gpt-oss 11%, Pj)	0.441 [0.429, 0.452]	0.35-0.70	held
Pj-P3-w1-g11	w1 time share (gpt-oss 11%, Pj)	0.173 [0.166, 0.181]	≤ 0.30	held
Pj-P4-w8r5-g11	w8r5 time share (gpt-oss 11%, Pj)	0.257 [0.247, 0.266]	0.15-0.45	held
Pj-P4-allr5-g11	allr5 time share (gpt-oss 11%, Pj)	0.482 [0.471, 0.494]	0.30-0.60	held
Pj-P5a-g11	MIN's set worth more with one read than two (gpt-oss 11%, Pj)	3.093 [2.977, 3.214]	> 0 ms	held
Pj-P5b-g11	single read within 5% of deployed (gpt-oss 11%, Pj)	0.998 [0.997, 0.999]	0.95-1.05	held
Pj-P6-fetch-g11	fetch/base (gpt-oss 11%, Pj)	1.265 [1.257, 1.273]	1.1-1.55	held
Pj-P6-both3p-g11	both3p/base (gpt-oss 11%, Pj)	1.428 [1.414, 1.442]	1.25-1.9	held
Pj-P6-order-g11	both3p $>$ fetch $>$ foa (gpt-oss 11%, Pj)	1.000	ordering	held (point)
Pj-P9-g11	host plan at most 150 μ s per step (gpt-oss 11%, Pj)	198.2	≤ 150 μ s	failed
Pj-P7-g11	aa / foa in [0.92, 1.04] on a ≥ 40 GB/s link (gpt-oss 11%, Pj)	0.880 [0.877, 0.882]	0.92-1.04	failed
Pj-P1-g25	engine reads within 4% of the replay (gpt-oss 25%, Pj)	0.016	≤ 0.04	held (point)
Pj-P3-w16-g25	w16 time share (gpt-oss 25%, Pj)	0.607 [0.600, 0.615]	0.45-0.85	held
Pj-P3-w4-g25	w4 time share (gpt-oss 25%, Pj)	0.244 [0.239, 0.250]	0.12-0.40	held
Pj-P3-w1-g25	w1 time share (gpt-oss 25%, Pj)	0.090 [0.077, 0.099]	≤ 0.30	held
Pj-P4-w8r5-g25	w8r5 time share (gpt-oss 25%, Pj)	0.151 [0.147, 0.156]	≤ 0.30	held
Pj-P4-allr5-g25	allr5 time share (gpt-oss 25%, Pj)	0.430 [0.420, 0.441]	0.30-0.60	held
Pj-P5a-g25	MIN's set worth more with one read than two (gpt-oss 25%, Pj)	0.971 [0.899, 1.040]	> 0 ms	held
Pj-P5b-g25	single read within 5% of deployed (gpt-oss 25%, Pj)	1.002 [0.999, 1.005]	0.95-1.05	held
Pj-P6-fetch-g25	fetch/base (gpt-oss 25%, Pj)	1.309 [1.296, 1.322]	1.1-1.55	held
Pj-P6-both3p-g25	both3p/base (gpt-oss 25%, Pj)	1.658 [1.634, 1.681]	1.25-1.9	held
Pj-P6-order-g25	both3p $>$ fetch $>$ foa (gpt-oss 25%, Pj)	1.000	ordering	held (point)
Pj-P9-g25	host plan at most 150 μ s per step (gpt-oss 25%, Pj)	193.3	≤ 150 μ s	failed
Pj-P7-g25	aa / foa in [0.92, 1.04] on a ≥ 40 GB/s link (gpt-oss 25%, Pj)	0.921 [0.917, 0.925]	0.92-1.04	held (point)
P2a	time share within 0.15 of read share at $\geq 80\%$ of hybrid host-cells	0.940	≥ 0.80	held (point)
P2b	w1 $<$ w4 $<$ w16 at $\geq 80\%$ of host-cells	1.000	≥ 0.80	held (point)
P8-g11	spread of fetch/base across hosts $>$ 3x the median within-host half-width (gpt-oss 11%)	55.5	> 3	held (point)
P8-g25	spread of fetch/base across hosts $>$ 3x the median within-host half-width (gpt-oss 25%)	36.5	> 3	held (point)